

Q1: If the kinetic energy of a moving body becomes four times its initial Kinetic energy, then the percentage change in its momentum will be:

- (A) 100%
- (B) 200%
- (C) 300%
- (D) 400%

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Solution:

Momentum, $P = \sqrt{2 km}$

$$\% \text{ change in } P = \frac{P_2 - P_1}{P_1} \times 100\%$$

$$= \frac{\sqrt{2k_2m} - \sqrt{2k_1m}}{\sqrt{2k_1m}} \times 100\%$$

$$= \frac{\sqrt{k_2} - \sqrt{k_1}}{\sqrt{k_1}} \times 100\%$$

$$= \left(\sqrt{\frac{k_2}{k_1}} - 1 \right) \times 100\%$$

$$= \left(\sqrt{\frac{4k_1}{k_1}} - 1 \right) \times 100\%$$

$$= 100\%$$

Q2: A boy reaches the airport and finds that the escalator is not working. He walks up the stationary escalator in time t_1 . If he remains stationary on a moving escalator then the escalator takes him up in time t_2 . The time taken by him to walk up on the moving escalator will be:

(A) $\frac{t_1 t_2}{t_2 - t_1}$

(B) $\frac{t_1 + t_2}{2}$

(C) $\frac{t_1 t_2}{t_2 + t_1}$

(D) $t_2 - t_1$

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(D) $t_2 - t_1$

Solution:

Let l be the length of escalator, v_1 be the velocity of man w.r.t. escalator, v_2 be the velocity of escalator.

$$t_1 = \frac{l}{v_1}$$

$$t_2 = \frac{l}{v_2}$$

$$t = \frac{l}{v_1 + v_2} = \frac{l}{\frac{l}{t_1} + \frac{l}{t_2}}$$

$$t = \frac{t_1 t_2}{t_1 + t_2}$$

Q3: A satellite is launched into a circular orbit of radius R around earth, while a second satellite is launched into a circular orbit of radius $1.02R$. the percentage difference in the time periods of the two satellites is:

- (A) 1.5
- (B) 2.0
- (C) 0.7
- (D) 3.0

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(B) 2.0

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(D) 3.0

Solution:

$$T^2 = CR^3$$

$$2\frac{dT}{T} = 3\frac{dR}{R}$$

$$\frac{dT}{T} = \frac{3}{2} \times \frac{0.02R}{R}$$

$$= 0.03$$

So % difference in the time period

$$= 0.03 \times 100 = 3\%$$

Q4: With what speed should a galaxy move outward with respect to earth so that the sodium-D line at wavelength $5890 \text{ \AA}$ is observed at $5896 \text{ \AA}$?

- (A) 306 km/sec
- (B) 322 km/sec
- (C) 296 km/sec
- (D) 336 km/sec

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Solution:

$$\lambda_{\text{obs}} = \lambda_{\text{actual}} \sqrt{\frac{1 + \frac{v}{c}}{1 - \frac{v}{c}}}$$
$$5896 = 5890 \left(1 + \frac{v}{c} \right) \text{ for } [V \ll C]$$
$$V = \frac{C \times 6}{5890} = 306 \text{ km/s}$$

Q5: The length of a metal wire is ℓ_1 , when the tension in it is T_1 and is ℓ_2 when the tension is T_2 . The natural length of the wire is:

(A) $\sqrt{\ell_1 \ell_2}$

(B) $\frac{\ell_1 T_2 - \ell_2 T_1}{T_2 - T_1}$

(C) $\frac{\ell_1 T_2 + \ell_2 T_1}{T_2 + T_1}$

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(D) $\frac{\ell_1 + \ell_2}{2}$

Solution:

$$T_1 = k(l_1 - l_0)$$

$$T_2 = k(l_2 - l_0)$$

From (i) and (ii)

$$l_0 = \frac{l_1 T_2 - l_2 T_1}{T_2 - T_1}$$

Q6: In an electromagnetic wave the electric field vector and magnetic field vector are given as $\vec{E} = E_0 \hat{i}$ and $\vec{B} = B_0 \hat{k}$ respectively. The direction of propagation of electromagnetic wave is along:

(A) $(\hat{k})$

(B) $\hat{j}$

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(A) $(\hat{k})$

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Solution:

$$\begin{aligned}\text{Direction of propagation of wave} &= \text{Direction of } \vec{E} \times \vec{B} \\ &= \text{Direction of } E_0 \hat{i} \times B_0 \hat{k} \\ &= \text{Direction of } E_0 B_0 (-\hat{j}) \\ &= -\hat{j}\end{aligned}$$

Q7: For a series LCR circuit with $R = 100\ \Omega$, $L = 0.5\text{ mH}$ and $C = 0.1\text{ pF}$ connected across $220\text{ V} - 50\text{ Hz}$ AC supply, the phase angle between current and supplied voltage and the nature of the circuit is:

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- (B) $\approx 90^\circ$, predominantly inductive
- (C) 0° , resonance circuit
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Solution:

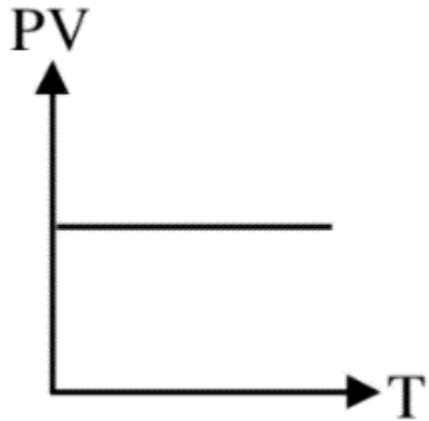
$$X_L = 2 \times \pi \times 50 \times 0.5 \times 10^{-3} \\ = 0.05\pi \, \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{10^{12}}{2\pi \times 50 \times 0.1} \\ = \frac{10^{11}}{\pi} \, \Omega$$

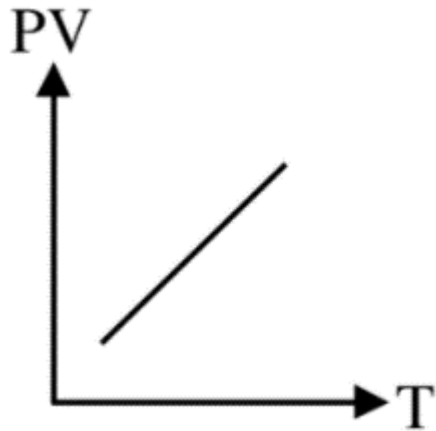
$$X_C - X_L \gg R$$

Q8: Which of the following graphs represent the behavior of an ideal gas? Symbols have their usual meaning.

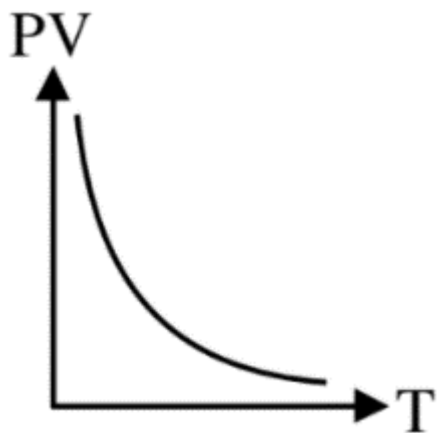
(A)



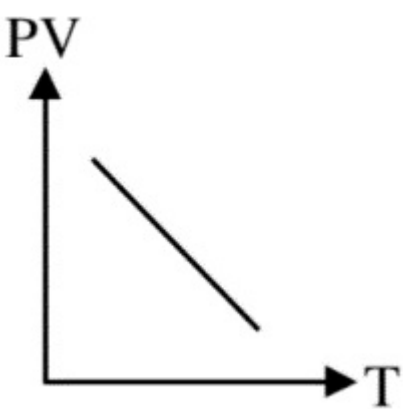
(B)



(C)

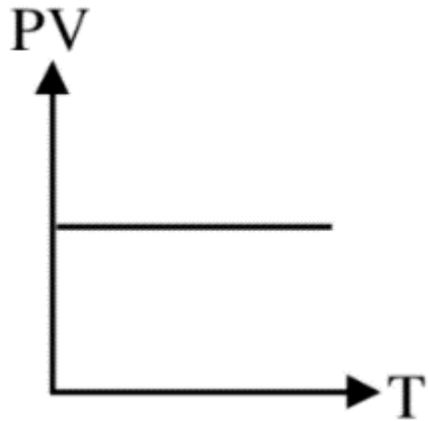


(D)

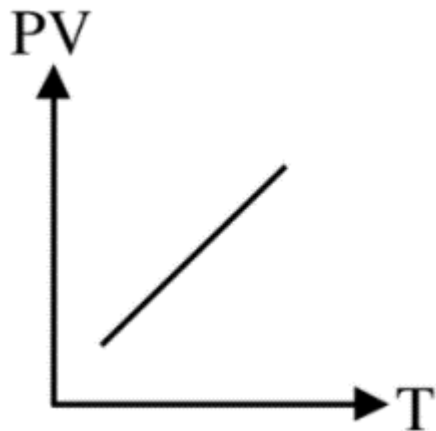


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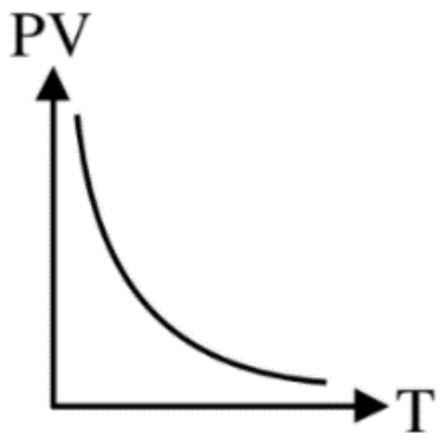
(A)



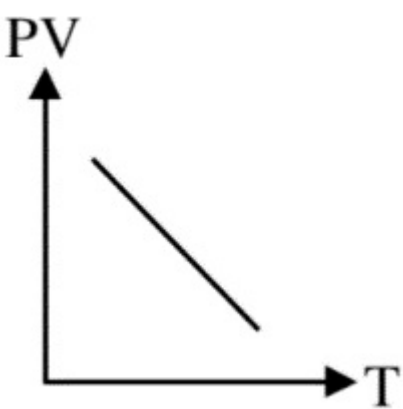
(B)



(C)



(D)



Solution:

$$PV=nRT \Rightarrow \mathbf{PV \propto T}$$

$\Rightarrow$ straight line with positive slope.

Q9: A particle is making simple harmonic motion along the X – axis. If at a distances x_1 *and* x_2 from the mean position the velocities of the particle are ν_1 *and* ν_2 respectively. The time period of its oscillation is given as:

(A)
$$T = 2\pi\sqrt{\frac{x_2^2+x_1^2}{\nu_1^2-\nu_2^2}}$$

(B)
$$T = 2\pi\sqrt{\frac{x_2^2+x_1^2}{\nu_1^2+\nu_2^2}}$$

(C)
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(C) $T = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{\nu_1^2 + \nu_2^2}}$

(D) $T = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{\nu_1^2 - \nu_2^2}}$

Solution:

$$\nu_1^2 = \omega^2 (A^2 - x_1^2)$$

$$\nu_2^2 = \omega^2 (A^2 - x_2^2)$$

$$\nu_1^2 - \nu_2^2 = \omega^2 (x_2^2 - x_1^2)$$

$$\nu_1^2 - \nu_2^2 = \left[\frac{2\pi}{T}\right]^2 (x_2^2 - x_1^2)$$

$$T = 2\pi\sqrt{\frac{(x_2^2 - x_1^2)}{(\nu_1^2 - \nu_2^2)}}$$

Q10: An electron having de-Broglie wavelength λ is incident on a target in a X – ray tube. Cut – off wavelength of emitted X – ray is:

(A) 0

(B) $\frac{2m^2c^2\lambda^2}{h^2}$

(C) $\frac{2mc\lambda^2}{h}$

(D) $\frac{hc}{mc}$

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Solution:

$$p = \frac{h}{\lambda}$$

$$\frac{p^2}{2m} = \frac{hc}{\lambda_0}$$

$$\frac{h^2}{2m\lambda^2} = \frac{hc}{\lambda_0}$$

$$\lambda_0 = \frac{2mc\lambda^2}{h}$$

Q11: A body rolls down an inclined plane without slipping. The kinetic energy of rotation is 50% of its translational kinetic energy. The body is:

- (A) Solid sphere
- (B) Solid cylinder
- (C) Hollow cylinder
- (D) Ring

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Solution:

$$\frac{k_T}{k_R} = \frac{k_T}{\frac{1}{2}k_T} = 2$$

$$\frac{MR^2}{I_{CM}} = 2$$

$$\Rightarrow I_{CM} = \frac{MR^2}{2}$$

Q12: If time (t), velocity (ν), and angular momentum (l) are taken as the fundamental units. Then the dimension of mass (m) in terms of t , ν and l is:

(A) $[t^{-1}\nu^1l^{-2}]$

(B) $[t^1\nu^1l^{-1}]$

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Solution:

$$l = [M^1 L^2 T^{-1}]$$

$$l = \mu^\alpha t^\beta \nu^\gamma$$

$$l = [M^1 L^2 T^{-1}]^\alpha [T]^\beta [L^1 T^{-1}]^\gamma$$

$$\alpha = 1$$

$$-\alpha + \beta - \gamma = 0$$

$$2\alpha + \gamma = 0$$

$$\gamma = -2$$

$$\beta = -1$$

Q13: The correct relation between the degrees of freedom f and the ratio of specific heat γ is:

(A) $f = \frac{2}{\gamma-1}$

(B) $f = \frac{2}{\gamma+1}$

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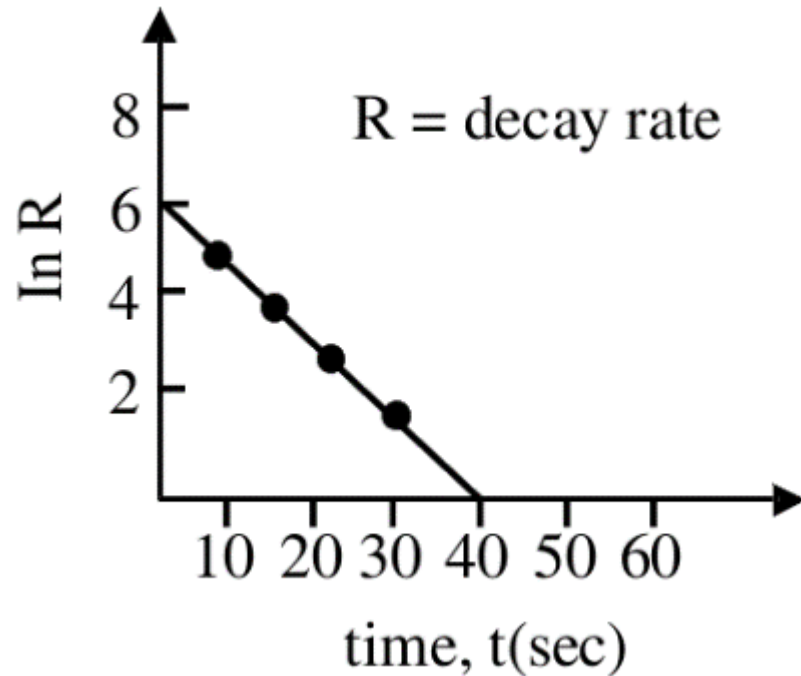
Solution:

$$\because \gamma = 1 + \frac{2}{f}$$

$$\Rightarrow \frac{2}{f} = \gamma - 1$$

$$\Rightarrow f = \frac{2}{\gamma-1}$$

Q14: For a certain radioactive process the graph between $\ln R$ and $t(\text{sec})$ is obtained as shown in the figure. Then the value of half life for the unknown radioactive is approximately:



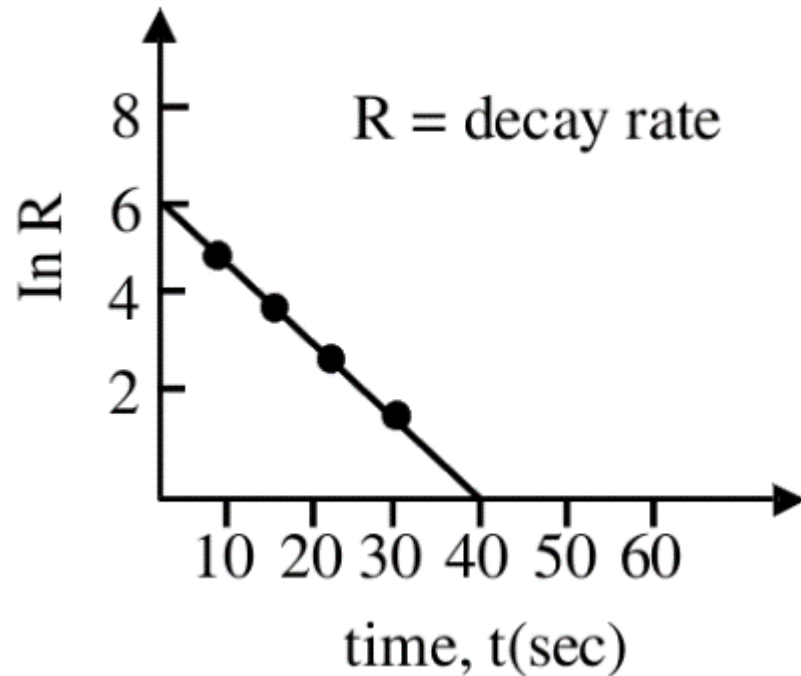
(A) 9.15 sec

(B) 6.93 sec

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Solution:

$$R = R_0 e^{-\lambda t}$$

$$\Rightarrow \ln R = \ln R_0 - \lambda t$$

$$\text{at } t = 0, \ln R = 6$$

$$\Rightarrow 6 = \ln R_0 \dots (1)$$

$$\text{at } t = 40 \text{ s, } \ln R = 0$$

$$\Rightarrow \lambda = \frac{\ln R_0}{t} = \frac{6}{40}$$

$$\Rightarrow t_{\frac{1}{2}} = \frac{\ln 2}{\lambda} = \frac{0.693 \times 40}{6} = 4.62 \text{ s}$$

Q15: Consider a binary star system of star A and star B with masses m_A and m_B , revolving in a circular orbit of radii r_A and r_B , respectively. If T_A and T_B are the time period of star A and star B, respectively, then:

(A) $\frac{T_A}{T_B} = \left(\frac{r_A}{r_B}\right)^{\frac{3}{2}}$

(B) $T_A = T_B$

(C) $T_A > T_B$ (if $m_A > m_B$)

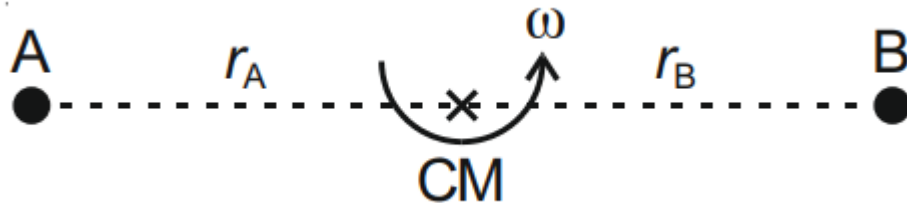
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- (C) $T_A > T_B$ (if $m_A > m_B$)
- (D) $T_A > T_B$ (if $r_A > r_B$)

Solution:

$$T_A = \frac{2\pi}{\omega}$$



$$T_B = \frac{2\pi}{\omega}$$
$$\Rightarrow T_A = T_B$$

Q16: At an angle of 30° to the magnetic meridian, the apparent dip is 45° . Find the true dip:

(A) $\tan^{-1} \sqrt{3}$

(B) $\tan^{-1} \frac{1}{\sqrt{3}}$

(C) $\tan^{-1} \frac{2}{\sqrt{3}}$

(D) $\tan^{-1} \frac{\sqrt{3}}{2}$

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Solution:

$$\tan \theta = \frac{B_V}{B_H}$$

$$\tan 45 = \frac{B_V}{B_H \cos 30} = \frac{2B_V}{B_H \sqrt{3}}$$

$$\Rightarrow B_V = \frac{\sqrt{3}}{2} B_H$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{\sqrt{3}}{2} \right)$$

Q17: A body at rest is moved along a horizontal straight line by a machine delivering a constant power. The distance moved by the body in time ' t ' is proportional to:

(A) $t^{\frac{3}{2}}$

(B) $t^{\frac{1}{2}}$

(C) $t^{\frac{1}{4}}$

(D) $t^{\frac{3}{4}}$

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Solution:

$$P = C$$

$$\text{So. K.E.} = Pt$$

$$\frac{1}{2}mv^2 = Pt$$

$$\Rightarrow v = \sqrt{\frac{2P}{m}}\sqrt{t}$$

$$\Rightarrow v = C\sqrt{t}$$

$$\frac{ds}{dt} = C\sqrt{t}$$

$$\int ds = \int C\sqrt{t}dt$$

$$\text{Distance} = Ct^{3/2}$$

Q18: Two vectors $\vec{P}$ and $\vec{Q}$ have equal magnitudes. If the magnitude of $\vec{P} + \vec{Q}$ is n times the magnitude of $\vec{P} - \vec{Q}$, then angle between $\vec{P}$ and $\vec{Q}$ is:

(A) $\sin^{-1} \left(\frac{n-1}{n+1} \right)$

(B) $\cos^{-1} \left(\frac{n-1}{n+1} \right)$

(C) $\sin^{-1} \left(\frac{n^2-1}{n^2+1} \right)$

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Solution:

$$\begin{aligned} |\vec{P} + \vec{Q}|^2 &= n^2 |\vec{P} - \vec{Q}|^2 \\ \Rightarrow P^2 + Q^2 + 2PQ \cos \theta &= n^2 (P^2 + Q^2 - 2PQ \cos \theta) \\ \Rightarrow 1 + 1 + 2 \cos \theta &= n^2 (1 + 1 - 2 \cos \theta) \\ \Rightarrow 1 + \cos \theta &= n^2 (1 - \cos \theta) \\ \Rightarrow \frac{1 + \cos \theta}{1 - \cos \theta} &= \frac{n^2}{1} \\ \Rightarrow \frac{1}{\cos \theta} &= \frac{n^2 + 1}{n^2 - 1} \\ \Rightarrow \theta &= \cos^{-1} \left(\frac{n^2 - 1}{n^2 + 1} \right) \end{aligned}$$

Q19: Two small drops of mercury each of radius R coalesce to form a single large drop. The ratio of total surface energy before and after the change is:

(A) $2^{\frac{1}{3}} : 1$

(B) $1 : 2^{\frac{1}{3}}$

(C) $2 : 1$

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Solution:

$$\frac{4}{3}\pi R^3 \times 2 = \frac{4}{3}\pi R_2^3$$

$$\Rightarrow R_2 = R \times 2^{\frac{1}{3}}$$

$$\therefore \frac{E_1}{E_2} = \frac{T \times 4\pi R^2 \times 2}{T \times 4\pi R_2^2 \times 2^{\frac{2}{3}}}$$

$$= \frac{2}{2^{2/3}} = 2^{\frac{1}{3}}$$

Q20: The magnetic susceptibility of a material of a rod is 499. Permeability in vacuum is $4\pi \times 10^{-7} \text{ H/m}$. Absolute permeability of the material of the rod is:

(A) $4\pi \times 10^{-4} \text{ H/m}$

(B) $2\pi \times 10^{-4} \text{ H/m}$

(C) $3\pi \times 10^{-4} \text{ H/m}$

(D) $\pi \times 10^{-4} \text{ H/m}$

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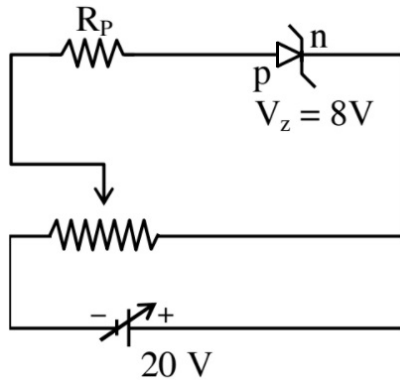
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Solution:

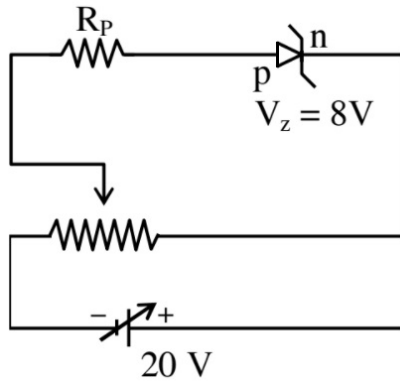
$$\chi = 499$$

$$\begin{aligned}\therefore \mu &= (1 + \chi) \times \mu_0 \\ &= 500 \times 4\pi \times 10^{-7} \\ &= 2\pi \times 10^{-4} \text{ H/m}\end{aligned}$$

Q21: A zener diode having zener voltage 8 V and power dissipation rating of 0.5 W is connected across a potential divider arranged with maximum potential drop across Zener diode is as shown in the diagram. The value of protective resistance *R_P is Ω .*



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192.00

Solution:

$$P = V_z I_z \Rightarrow I_z = \frac{1}{16} \text{ A}$$

$$I_z = \frac{V - V_z}{R_P}$$

$$R_P = (20 - 8) \times 16 = 192 \Omega$$

Q22: A body of mass ' m ' is launched up on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of friction between the body and plane is $\frac{\sqrt{x}}{5}$ if the time of ascent is half of the time of descent, the value of x is _____.

Q22: A body of mass ' m ' is launched up on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of friction between the body and plane is $\frac{\sqrt{x}}{5}$ if the time of ascent is half of the time of descent, the value of x is _____.

3.00

Solution:

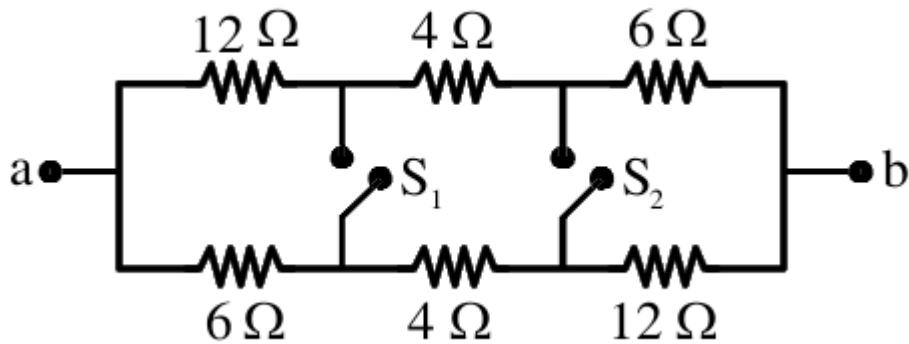
$$t_A = \sqrt{\frac{2\ell}{g(\sin \theta + \mu \cos \theta)}}$$

$$t_D = \sqrt{\frac{2\ell}{g(\sin \theta - \mu \cos \theta)}}$$

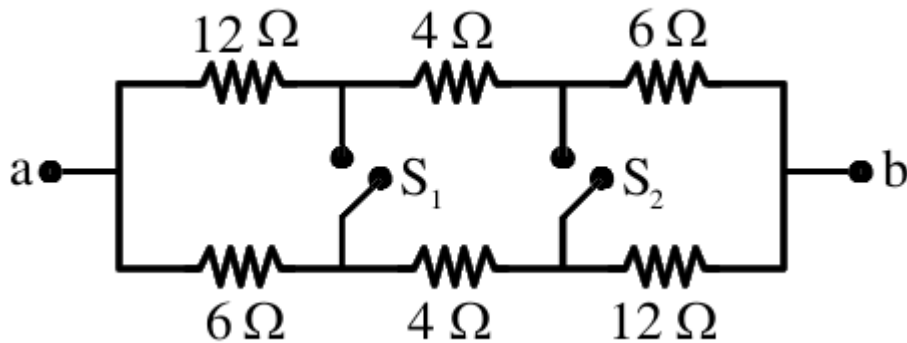
$$t_A = \frac{1}{2} t_D$$

$$\Rightarrow \mu = \frac{3}{5} \tan \theta = \frac{\sqrt{3}}{5}$$

Q23: In the given figure switches S_1 and S_2 are in open condition. The resistance across ab when the switches S_1 and S_2 are closed is _____ Ω .



Q23: In the given figure switches S_1 and S_2 are in open condition. The resistance across ab when the switches S_1 and S_2 are closed is _____ Ω .



10.00

Solution:

$$R_{ab} = \frac{12 \times 6}{12 + 6} + \frac{4 \times 4}{4 + 4} + \frac{6 \times 12}{6 + 12}$$

$$= 10\Omega$$

Q24: Two bodies, a ring and a solid cylinder of same material are rolling down without slipping an inclined plane. The radii of the bodies are same. The ratio of velocity of the centre of mass at the bottom of the inclined plane of the ring to that of the cylinder is $\frac{\sqrt{x}}{2}$. Then, the value of x is _____.

Q24: Two bodies, a ring and a solid cylinder of same material are rolling down without slipping an inclined plane. The radii of the bodies are same. The ratio of velocity of the centre of mass at the bottom of the inclined plane of the ring to that of the cylinder is $\frac{\sqrt{x}}{2}$. Then, the value of x is _____.

3.00

Solution:

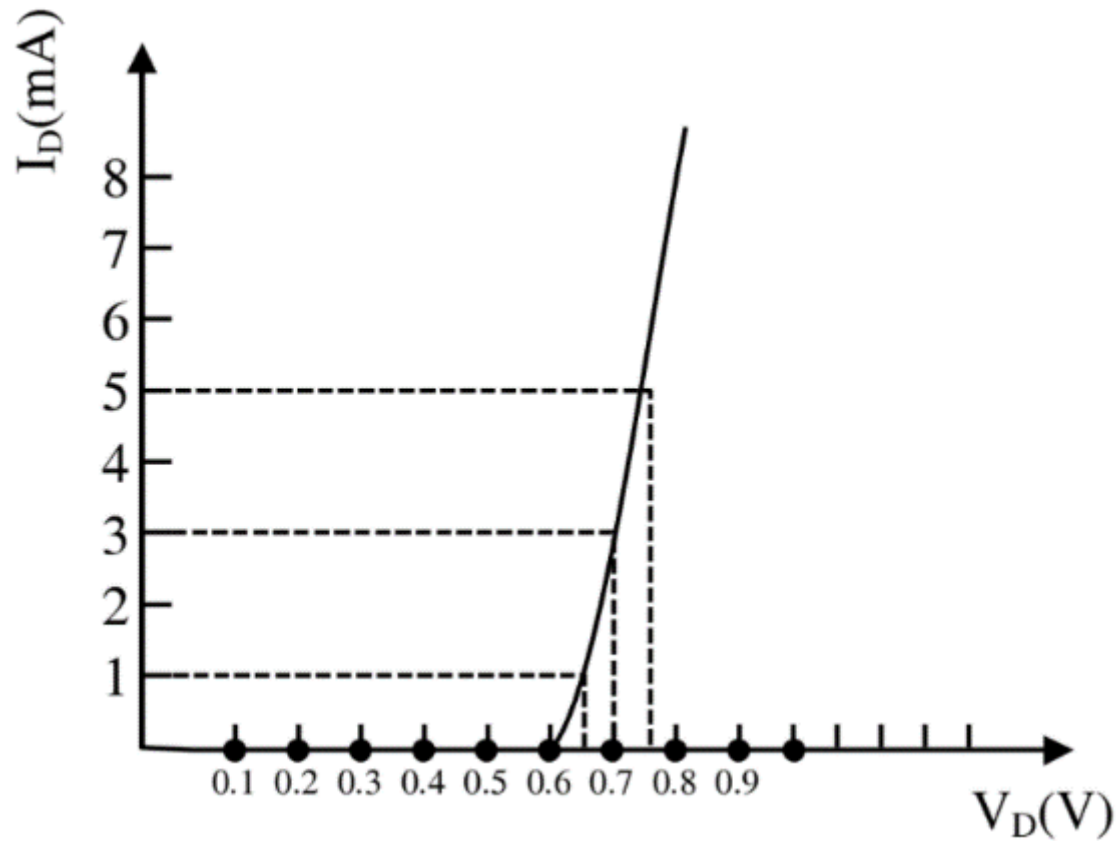
$$v = \sqrt{\frac{2gh}{1 + \frac{K^2}{R^2}}}$$

$$v_{\text{ring}} = \sqrt{gh}$$

$$v_{\text{cylinder}} = \sqrt{\frac{4}{3}gh}$$

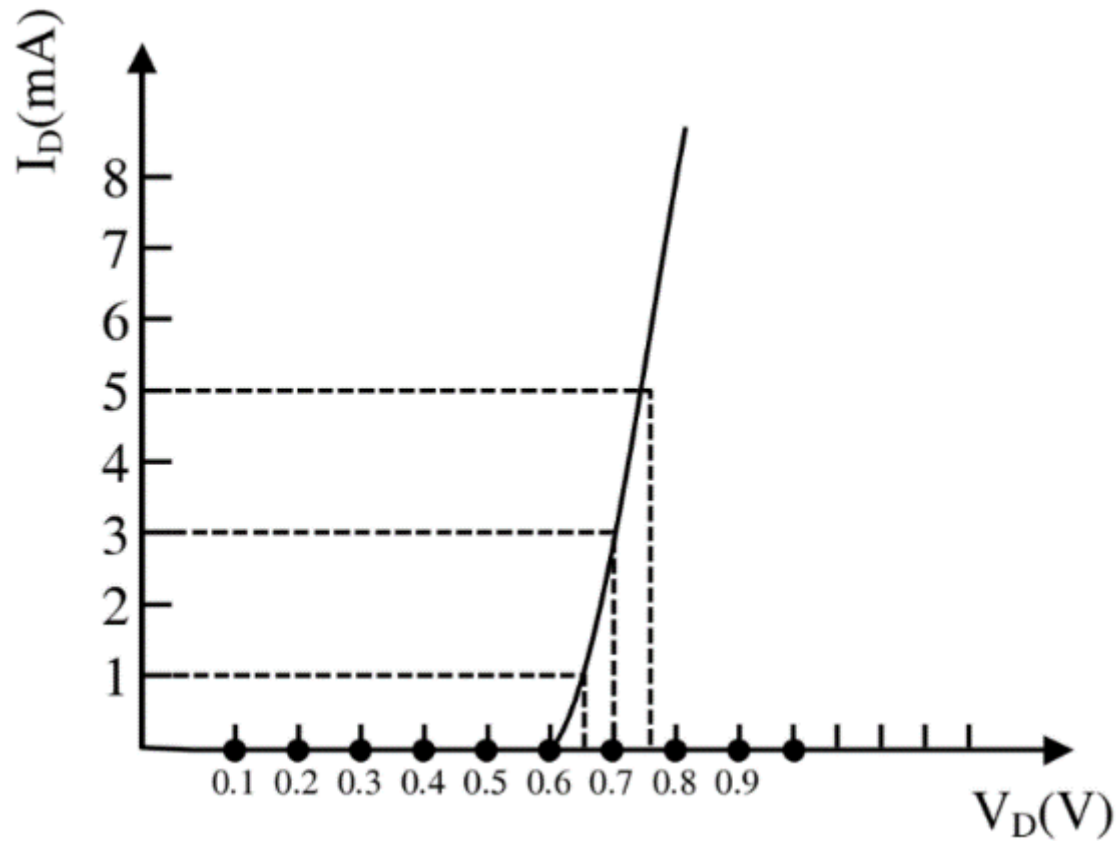
$$\frac{v_{\text{ring}}}{v_{\text{cylinder}}} = \sqrt{\frac{3}{4}}$$

Q25: For the forward biased diode characteristics shown in the figure, the dynamic resistance at $I_D = 3 \text{ mA}$ will be _____ Ω .





Q25: For the forward biased diode characteristics shown in the figure, the dynamic resistance at $I_D = 3 \text{ mA}$ will be _____ Ω .



25.00

Solution:

$$R_D = \frac{\Delta V}{\Delta I} = \frac{0.1}{4 \times 10^{-3}} = 25 \Omega$$

Q26: A series LCR circuit of $R = 5\Omega$, $L = 20 \text{ mH}$ and $C = 0.5 \mu\text{F}$ is connected across an AC supply of 250 V, having variable frequency. The power dissipated at resonance condition is _____ $\times 10^2 \text{ W}$.

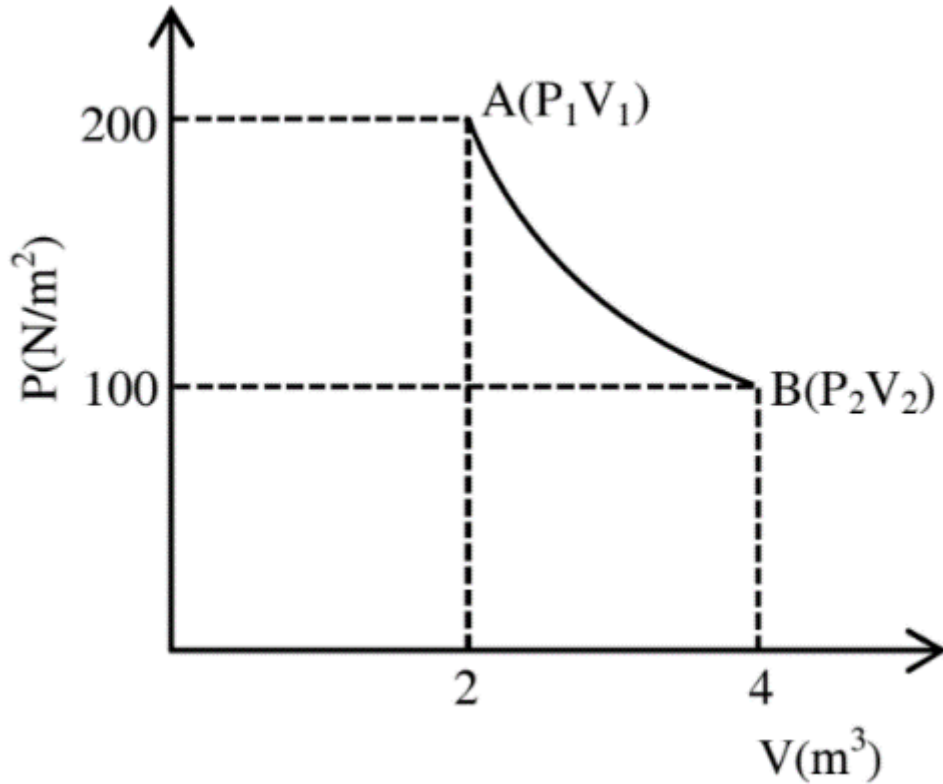
Q26: A series LCR circuit of $R = 5\Omega$, $L = 20\text{ mH}$ and $C = 0.5\text{ }\mu\text{F}$ is connected across an AC supply of 250 V, having variable frequency. The power dissipated at resonance condition is _____ $\times 10^2\text{ W}$.

125.00

Solution:

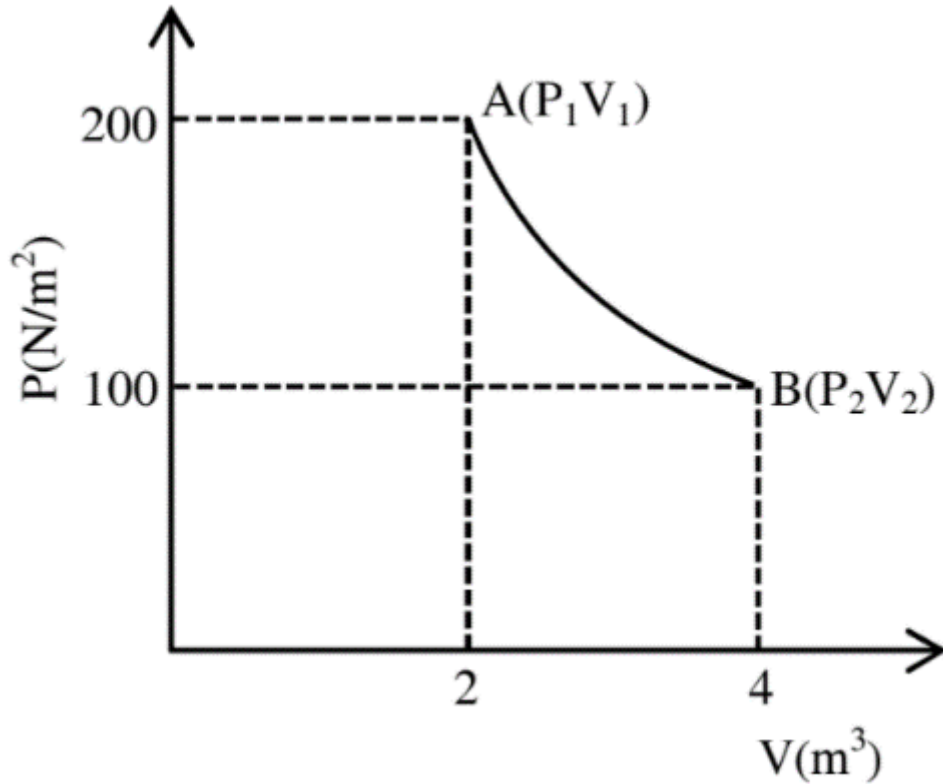
$$\begin{aligned} P_r &= \frac{V_{\text{rms}}^2}{R} \text{ as at resonance } Z=R \\ &= \frac{250 \times 250}{5} = 125 \times 10^2 \text{ W} \end{aligned}$$

Q27: One mole of an ideal gas at 27°C is taken from A to B as shown in the given PV indicator diagram. The work done by the system will be $\text{_____} \times 10^{-1} \text{ J}$.
[Given : $R = 8.3 \text{ J/mole K}$, $\ln 2 = 0.6931$]





Q27: One mole of an ideal gas at 27°C is taken from A to B as shown in the given PV indicator diagram. The work done by the system will be $\text{_____} \times 10^{-1} \text{ J}$.
[Given : $R = 8.3 \text{ J/mole K}$, $\ln 2 = 0.6931$]



17258.00

Solution:

Assuming process to be isothermal

$$\begin{aligned} W &= nRT \ln\left(\frac{V_f}{V_i}\right) \\ &= 8.3 \times 300 \times \ln(2) \\ &= 1725.819 \text{ J} \\ &= 17258 \times 10^{-1} \text{ J} \end{aligned}$$

Q28: A certain metallic surface is illuminated by monochromatic radiation of wavelength λ . The stopping potential for photoelectric current for this radiation is $3V_0$. If the same surface is illuminated with a radiation of wavelength 2λ , the stopping potential is V_0 . The threshold wavelength of this surface for photoelectric effect is _____ λ .

Q28: A certain metallic surface is illuminated by monochromatic radiation of wavelength λ . The stopping potential for photoelectric current for this radiation is $3V_0$. If the same surface is illuminated with a radiation of wavelength 2λ , the stopping potential is V_0 . The threshold wavelength of this surface for photoelectric effect is _____ λ .

Solution:

$$e3V_0 = \frac{hc}{\lambda} - \frac{hc}{\lambda_0} \dots (i)$$

$$eV_0 = \frac{hc}{2\lambda} - \frac{hc}{\lambda_0} \dots (ii)$$

$$\Rightarrow \lambda_0 = 4\lambda$$

Q29: A body rotating with an angular speed of 600 rpm is uniformly accelerated to 1800 rpm in 10 sec. the number of rotations made in the process is _____.

Q29: A body rotating with an angular speed of 600 rpm is uniformly accelerated to 1800 rpm in 10 sec. the number of rotations made in the process is _____.

200.00

Solution:

$$4\pi^2 \frac{(18^2 - 6^2)10^4}{60^2} = \frac{2 \times 1200}{10} \times \frac{\pi^2 \times \theta}{60}$$

$$\theta = \frac{\pi \times 24 \times 12 \times 10^4}{120 \times 60} = \frac{\pi 24 \times 10^3}{60}$$

$$\theta = \frac{(12000)(2\pi)}{60}$$

$$= (200)2\pi$$

Q30: A radioactive substance decays to $\left(\frac{1}{16}\right)^{th}$ of its initial activity in 80 days. The half life of the radioactive substance expressed in days is _____.

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Solution:

$$\frac{x}{16} = x\left(\frac{1}{2}\right)^n$$

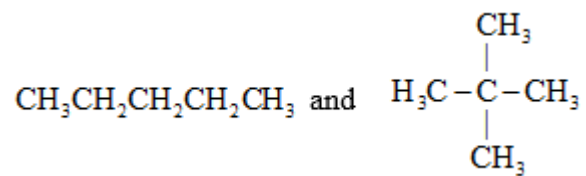
$$n = 4$$

$$\Rightarrow 4 \text{ half lives} = 80 \text{ days}$$

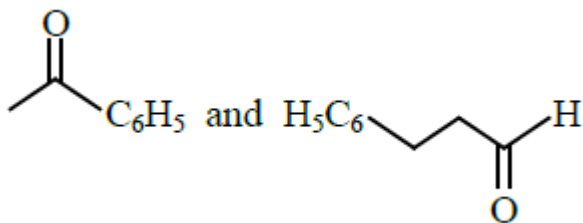
$$1 \text{ half life} = 20 \text{ days}$$

Q31: Which one of the following pairs of isomers is an example of metamerism?

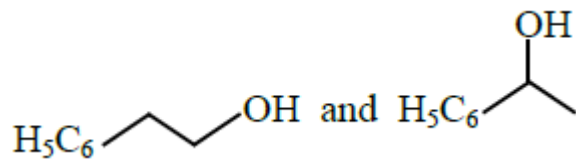
(A)



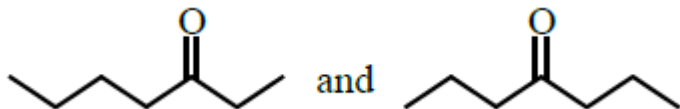
(B)



(C)

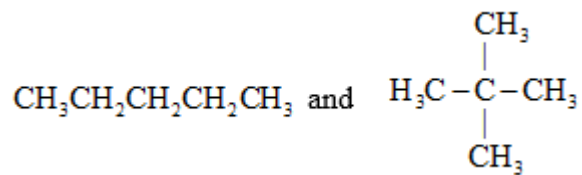


(D)

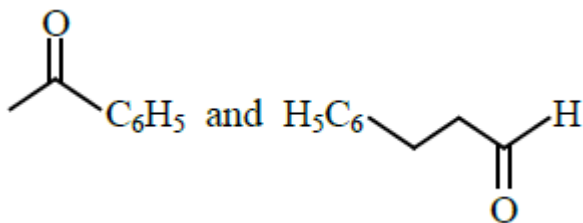


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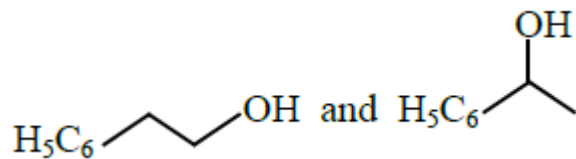
(A)



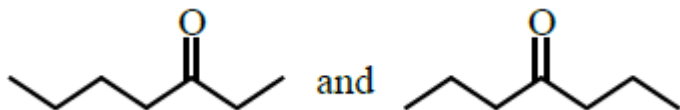
(B)



(C)

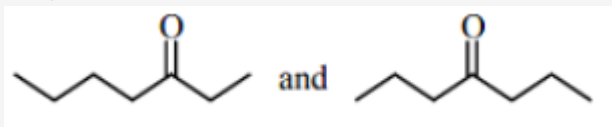


(D)

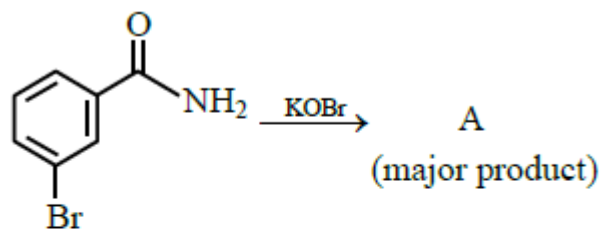


Solution:

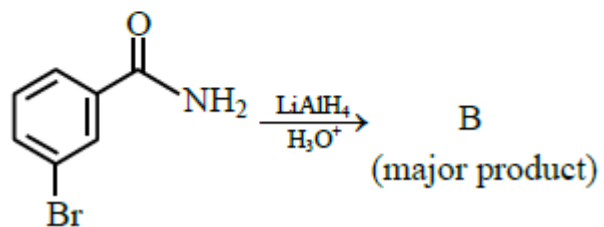
Metamers are compounds which have different alkyl groups present along both side of polyvalent functional group.
Only options 4 represent metamers



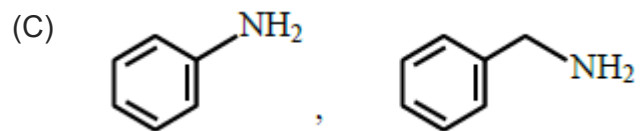
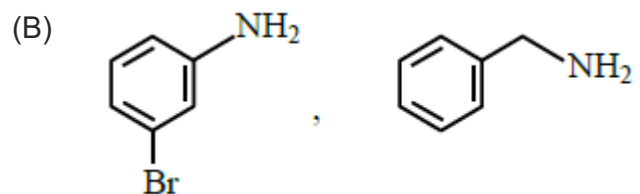
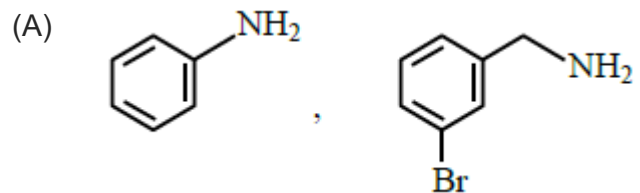
- (1) These are chain isomers
- (2) Not isomers having different molecular formula
- (3) Position isomers



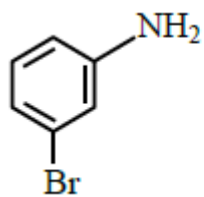
Q32:



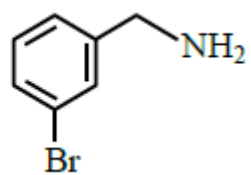
In the above reactions, product A and product B respectively are:

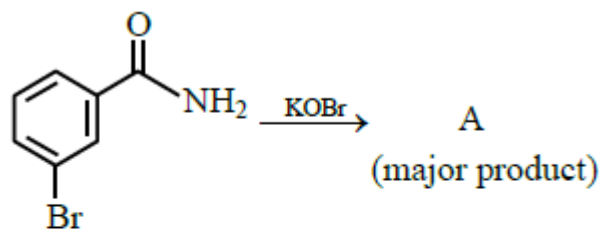


(D)

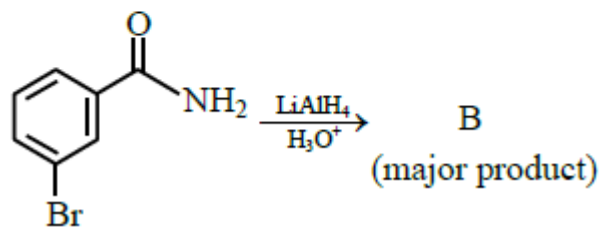


,

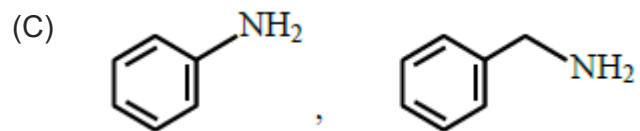
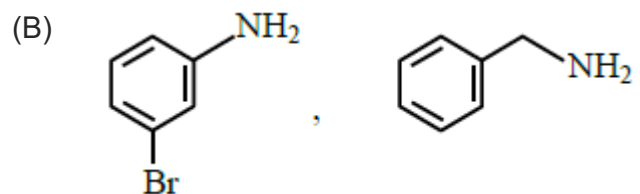
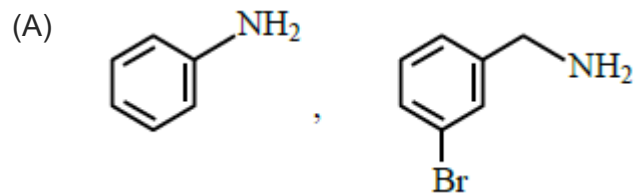




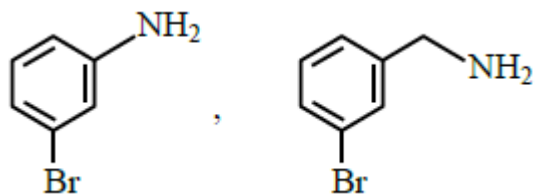
Q32:



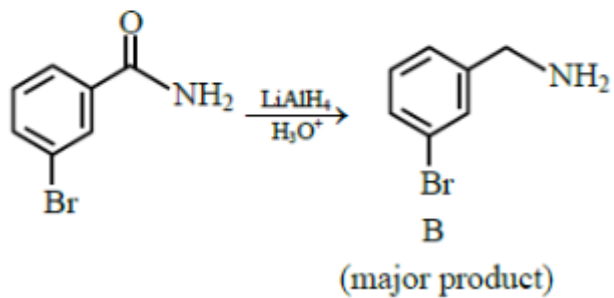
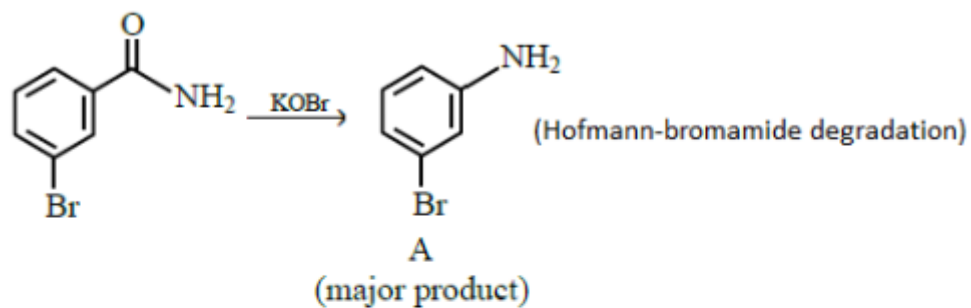
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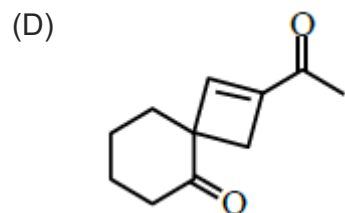
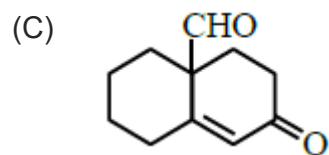
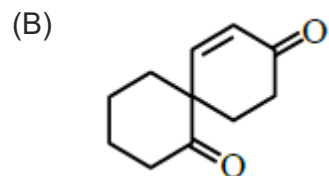
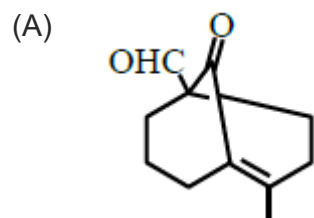
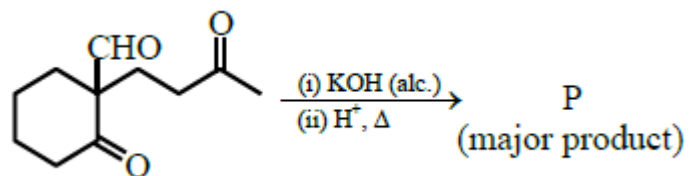
(D)



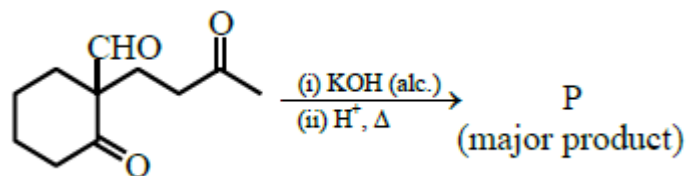
Solution:



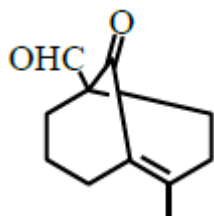
Q33: The major product (P) in the following reaction is:



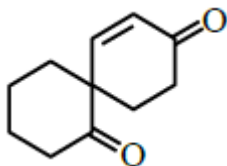
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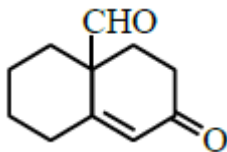
(A)



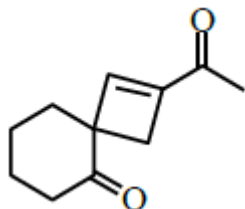
(B)



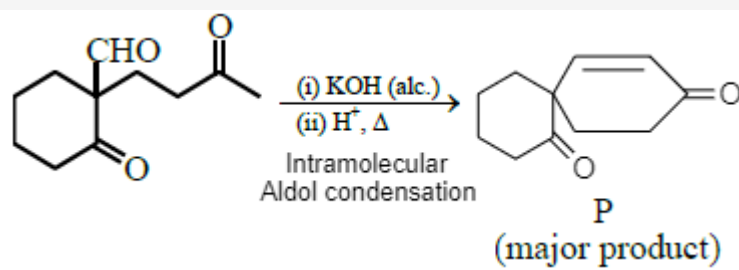
(C)



(D)



Solution:



Q34: The single largest industrial application of dihydrogen is:

- (A) Manufacture of metal hydrides
- (B) Rocket fuel in space research
- (C) In the synthesis of ammonia
- (D) In the synthesis of nitric acid

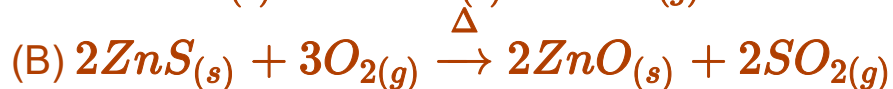
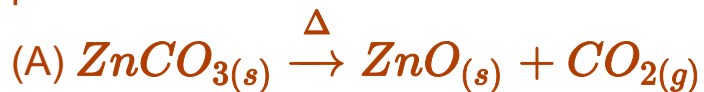
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- (A) Manufacture of metal hydrides
- (B) Rocket fuel in space research
- (C) In the synthesis of ammonia
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Solution:

The largest single use of dihydrogen is in the synthesis of ammonia which is used in the manufacture of nitric acid and nitrogenous fertilizers.

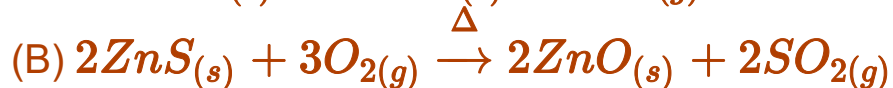
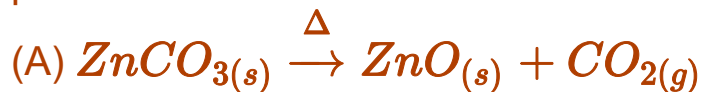
Q35: Consider two chemical reactions (A) and (B) that take place during metallurgical process:



The **correct** option of names given to them respectively is:

- (A) (A) is calcination and (B) is roasting
- (B) Both (A) and (B) are producing same product so both are roasting
- (C) Both (A) and (B) are producing same product so both are calcination
- (D) (A) is roasting and (B) is calcination

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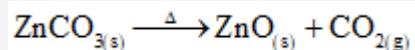


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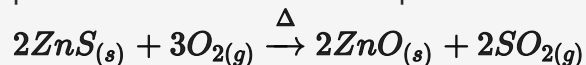
- (A) (A) is calcination and (B) is roasting
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- (C) Both (A) and (B) are producing same product so both are calcination
- (D) (A) is roasting and (B) is calcination

Solution:

Calcination is the process in which the concentrated ore is heated to a high temperature (just below its fusion temperature) in the absence of air (or limited supply of air). This process is mainly used for a carbonate ore to get its oxide.



Roasting is the process in which the concentrated ore is heated to a high temperature (just below its fusion temperature) in presence of excess of air. This process is mainly applicable for sulphide ores to get the corresponding metal oxides.



Q36: A solution is 0.1 M in Cl^- and 0.001 M in CrO_4^{2-} . Solid $AgNO_3$ is gradually added to it. Assuming that the addition does not change in volume and

$$K_{sp}(AgCl) = 1.7 \times 10^{-10} M^2 \text{ and } K_{sp}(Ag_2CrO_4) = 1.9 \times 10^{-12} M^3.$$

Select **correct** statement from the following:

- (A) $AgCl$ precipitates first because its K_{sp} is high
- (B) Ag_2CrO_4 precipitates first as its K_{sp} is low
- (C) Ag_2CrO_4 precipitates first because the amount of Ag^+ needed is low
- (D) $AgCl$ will precipitates first as the amount of Ag^+ needed to precipitate is low

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- (D) $AgCl$ will precipitates first as the amount of Ag^+ needed to precipitate is low

Solution:

Given - A solution is 0.1 M in Cl^- and 0.001 M in CrO_4^{2-} .

$$\text{The amount of } Ag^+ \text{ just required to start precipitation of } AgCl = \frac{K_{sp}}{[Cl^-]} = \frac{1.7 \times 10^{-10}}{0.1} = 1.7 \times 10^{-9}$$

$$\text{The amount of } Ag^+ \text{ just required to start precipitation of } Ag_2CrO_4 = \sqrt{\frac{K_{sp}}{[CrO_4^{2-}]}} = \sqrt{\frac{1.9 \times 10^{-12}}{0.001}} = \sqrt{1.9 \times 10^{-9}}$$

Thus, $AgCl$ required less amount of Ag^+ to start precipitating, it will precipitate first.

Q37: Outermost electronic configuration of a group 13 element, E, is $4s^2, 4p^1$. The electronic configuration of an element of p-block period-five placed diagonally to element, E is

- (A) $[\text{Kr}]3d^{10}4s^24p^2$
- (B) $[\text{Ar}]3d^{10}4s^24p^2$
- (C) $[\text{Xe}]5d^{10}6s^26p^2$
- (D) $[\text{Kr}]4d^{10}5s^25p^2$

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- (D) $[\text{Kr}]4d^{10}5s^25p^2$

Solution:

	13 th	14 th	15 th
2 nd Period	B ($2s^22p^1$)	C	N
3 rd Period	Al ($3s^23p^1$)	Si	P
4 th Period	Ga ($4s^24p^1$)	Ge	As
5 th Period	In ($5s^25p^1$)	Sn	Sb

That element is Sn $\Rightarrow [\text{Kr}] 4d^{10} 5s^2 5p^2$

Q38: Metallic sodium does not react normally with:

- (A) gaseous ammonia
- (B) But-2-yne
- (C) Ethyne
- (D) tert-butyl alcohol

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- (A) gaseous ammonia
- ☒ (B) But-2-yne
- (C) Ethyne
- (D) tert-butyl alcohol

Solution:

Metallic sodium does not react with 2-butyne because 2-butyne does not have acidic hydrogen.

Q39: Spin only magnetic moment of an octahedral complex of Fe^{2+} in the presence of a strong field ligand in BM is:

- (A) 4.89
- (B) 2.82
- (C) 0
- (D) 3.46

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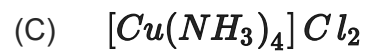
Solution:



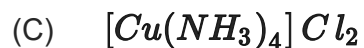
Unpaired e^- [$n = 0$]

$$\mu = \sqrt{n(n + 2)} = 0$$

Q40: Which one of the following species **doesn't** have a magnetic moment of 1.73 BM, (spin only value)?



Q40: Which one of the following species **doesn't** have a magnetic moment of 1.73 BM, (spin only value)?



Solution:

$\mu = 1.73$ BM it means number of unpaired electron = 1

Species	Unpaired electron
O_2^-	1
O_2^+	1
Cu^+	0
Cu^{2+}	1

Q41: Which one of the following statements is not true about enzymes?

- (A) Enzymes are non-specific for a reaction and substrate.
- (B) Almost all enzymes are proteins.
- (C) Enzymes work as catalysts by lowering the activation energy of a biochemical reaction.
- (D) The action of enzymes is temperature and pH specific

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- (D) The action of enzymes is temperature and pH specific

Solution:

- (a) Enzymes are highly specific in nature. Each enzyme is specific for a given reaction, i.e., one catalyst cannot catalyse more than one reaction. For example, the enzyme urease catalyses the hydrolysis of urea only. It does not catalyse hydrolysis of any other amide.
- (b) Enzymes are complex nitrogenous organic compounds which are produced by living plants and animals. They are actually protein molecules of high molecular mass and form colloidal solutions in water.
- (c) Enzymes work as catalysts by lowering the activation energy of a biochemical reaction.
- (d) The rate of an enzyme reaction becomes maximum at a definite temperature, called the optimum temperature. The rate of an enzyme-catalysed reaction is maximum at a particular pH called optimum pH, which is between pH values 5-7.

Q42: The hybridizations of the atomic orbitals of nitrogen in NO_2^- , NO_2^+ and NH_4^+ respectively are.

- (A) sp^3 , sp^2 and sp
- (B) sp, sp^2 and sp^3
- (C) sp^3 , sp and sp^2
- (D) sp^2 , sp and sp^3

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- (A) sp^3 , sp^2 and sp
- (B) sp, sp^2 and sp^3
- (C) sp^3 , sp and sp^2
- (D) sp^2 , sp and sp^3

Solution:

Species	Number of hybrid orbitals	Hybridization
NO_2^-	$\frac{1}{2}(5 + 1) = 3$	sp^2
NO_2^+	$\frac{1}{2}(5 - 1) = 2$	sp
NH_4^+	$\frac{1}{2}(5 + 4 - 1) = 4$	sp^3

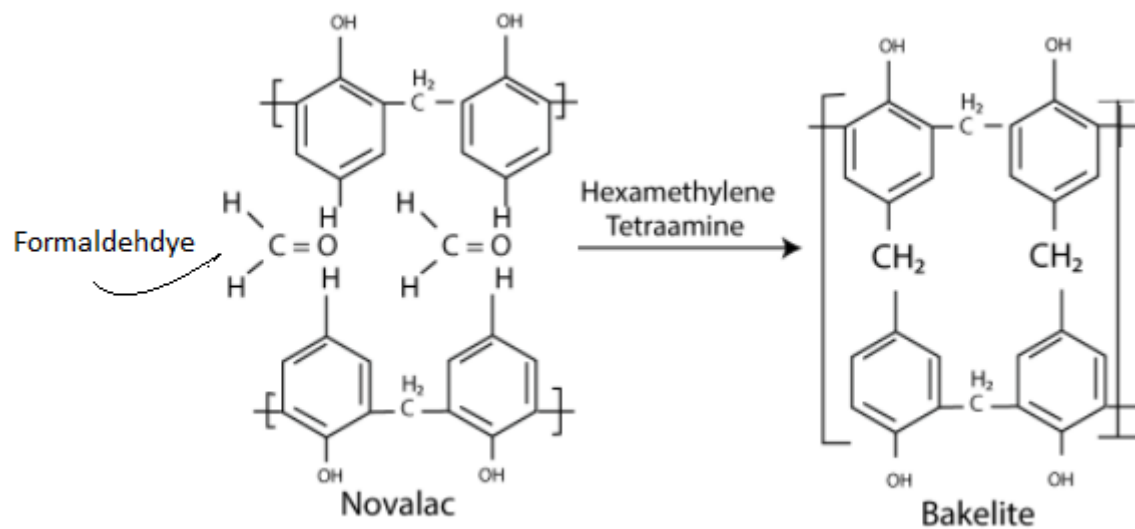
Q43: Bakelite is a cross-linked polymer of formaldehyde and:

- (A) PHBV
- (B) Buna-S
- (C) Novolac
- (D) Dacron

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- (A) PHBV
- (B) Buna-S
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- (D) Dacron

Solution:



Q44: Benzene on nitration gives nitrobenzene in presence of HNO_3 and H_2SO_4 mixture, where:

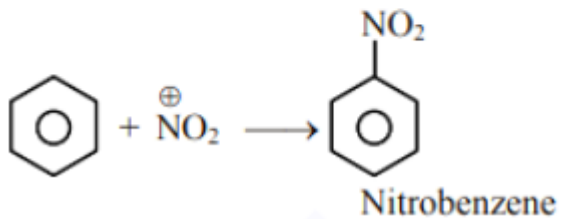
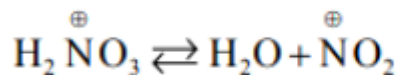
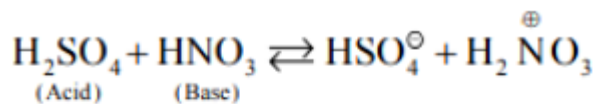
- (A) both H_2SO_4 and HNO_3 act as a bases
- (B) HNO_3 acts as an acid and H_2SO_4 acts as a base
- (C) both H_2SO_4 and HNO_3 act as an acids
- (D) HNO_3 acts as a base and H_2SO_4 acts as an acid

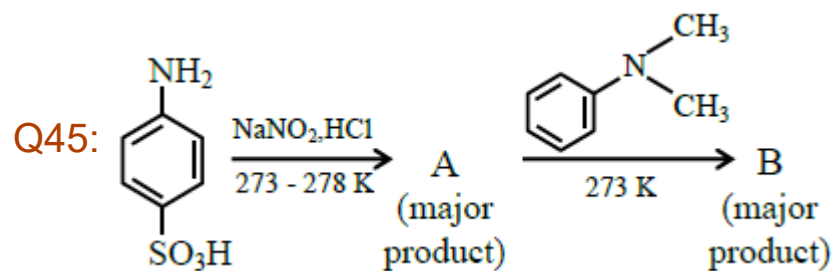
Q44: Benzene on nitration gives nitrobenzene in presence of HNO_3 and H_2SO_4 mixture, where:

- (A) both H_2SO_4 and HNO_3 act as a bases
- (B) HNO_3 acts as an acid and H_2SO_4 acts as a base
- (C) both H_2SO_4 and HNO_3 act as an acids
- (D) HNO_3 acts as a base and H_2SO_4 acts as an acid

Solution:

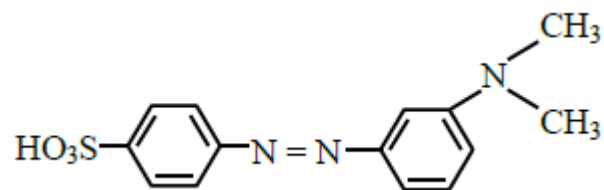
HNO_3 acts as a base and H_2SO_4 acts as an acid Reagent for nitration of Benzene.



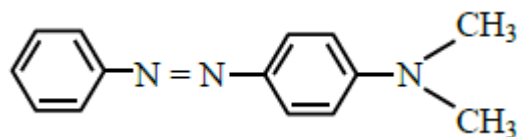


Consider the above reaction, compound B is:

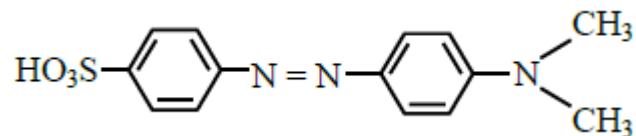
(A)



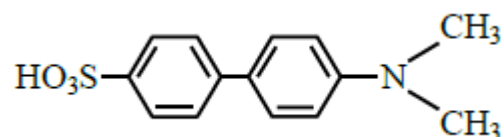
(B)

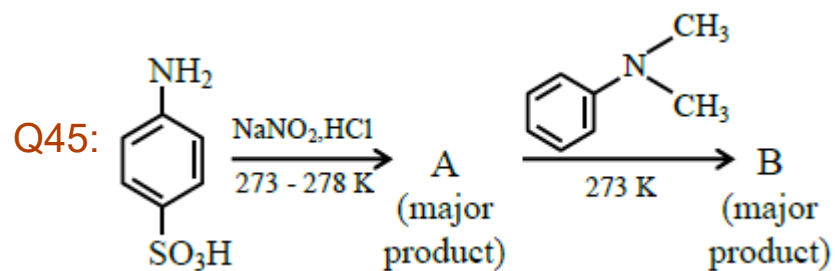


(C)



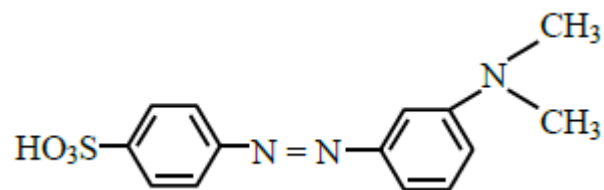
(D)



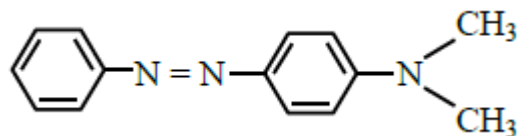


Consider the above reaction, compound B is:

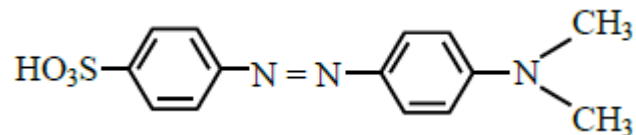
(A)



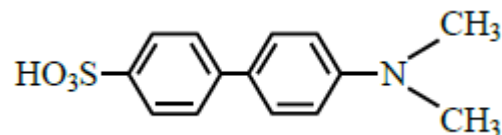
(B)



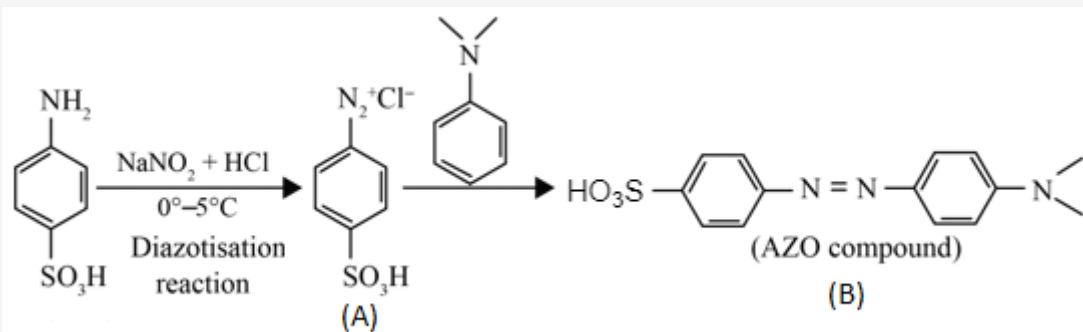
(C)

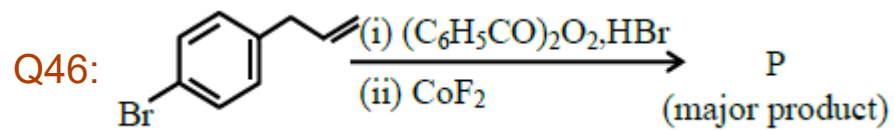


(D)

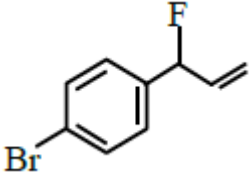
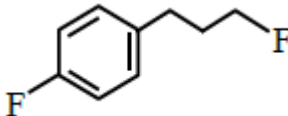
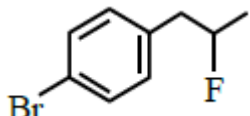
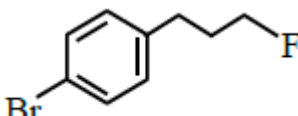


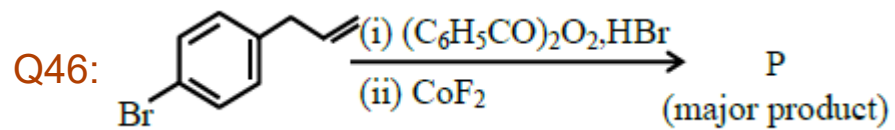
Solution:



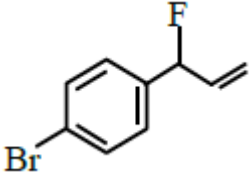
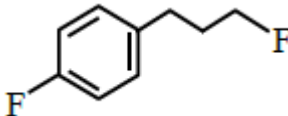
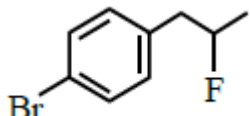
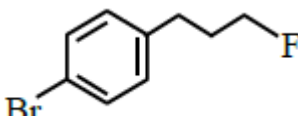


Major product P of above reaction, is:

- (A) 
- (B) 
- (C) 
- (D) 

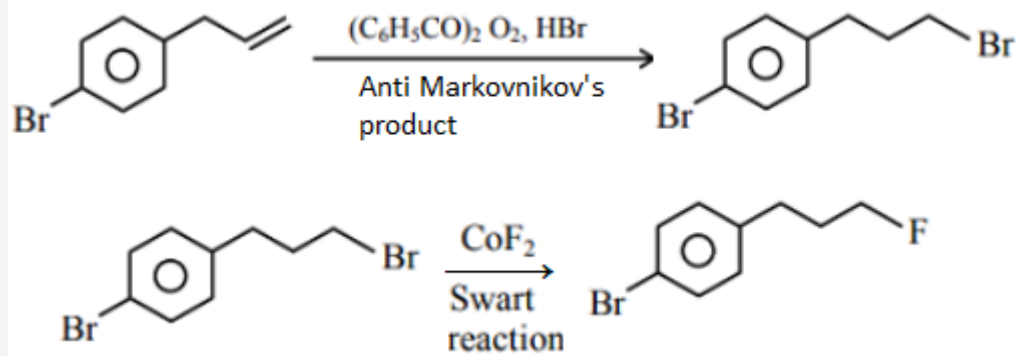


Major product P of above reaction, is:

- (A) 
- (B) 
- (C) 
- (D) 

Solution:

Here addition of HBr in presence of peroxide takes place according to anti-markovnikov rule.



Q47: Cu^{2+} salt reacts with potassium iodide to give

- (A) Cu_2I_2
- (B) Cu_2I_3
- (C) CuI
- (D) $\text{Cu}(\text{I}_3)_2$

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- (A) Cu_2I_2
- (B) Cu_2I_3
- (C) CuI
- (D) $\text{Cu}(\text{I}_3)_2$

Solution:

Cu^{2+} undergoes redox reaction.



Q48: In Carius method, halogen containing organic compound is heated with fuming nitric acid in the presence of:

- (A) HNO_3
- (B) $AgNO_3$
- (C) $CuSO_4$
- (D) $BaSO_4$

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- (A) HNO_3
- (B) $AgNO_3$
- (C) $CuSO_4$
- (D) $BaSO_4$

Solution:

Organic compound is heated with fuming nitric acid in the presence of silver nitrate in carius method. $AgNO_3$ is used as reagent to distinguish Cl^- , Br^- and I^- respectively.

Q49: Which one of the following gases is reported to retard photosynthesis?

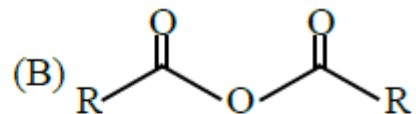
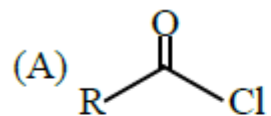
- (A) CO
- (B) CFCs
- (C) CO_2
- (D) NO_2

Q49: Which one of the following gases is reported to retard photosynthesis?

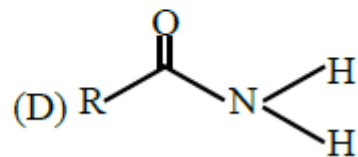
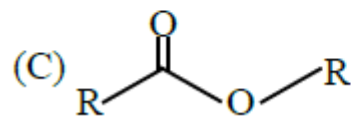
- (A) CO
- (B) CFCs
- (C) CO_2
- (D) NO_2

Solution:

Reason: NO_2 damage the leaves of plants and retard the photosynthesis.



Q50:



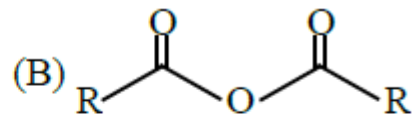
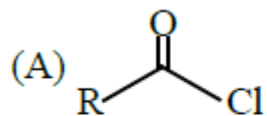
The **correct** order of their reactivity towards hydrolysis at room temperature is:

(A) (A) > (B) > (C) > (D)

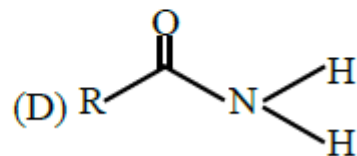
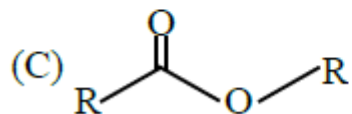
(B) (D) > (A) > (B) > (C)

(C) (D) > (B) > (A) > (C)

(D) (A) > (C) > (B) > (D)



Q50:



The **correct** order of their reactivity towards hydrolysis at room temperature is:

(A) (A) > (B) > (C) > (D)

(B) (D) > (A) > (B) > (C)

(C) (D) > (B) > (A) > (C)

(D) (A) > (C) > (B) > (D)

Solution:

Rate of hydrolysis is directly proportional to δ positive charge present on carbon of C = O group. Rate of hydrolysis follows following order: Acid chloride > Acid anhydride > ester > Acid amide

Q51: For a given chemical reaction $A \rightarrow B$ at 300 K the free energy change is $-49.4 \text{ kJ mol}^{-1}$ and the enthalpy of reaction is 51.4 kJ mol^{-1} . The entropy change of the reaction is _____ $\text{J K}^{-1} \text{ mol}^{-1}$.

Q51: For a given chemical reaction $A \rightarrow B$ at 300 K the free energy change is $-49.4 \text{ kJ mol}^{-1}$ and the enthalpy of reaction is 51.4 kJ mol^{-1} . The entropy change of the reaction is _____ $\text{J K}^{-1} \text{ mol}^{-1}$.

336.00

Solution:

$$\Delta G = \Delta H - T\Delta S$$

$$\Rightarrow -49400 = 51400 - 300\Delta S$$

$$\Rightarrow 300 \Delta S = 51400 + 49400 = 1,00,800$$

$$\Rightarrow \Delta S = \frac{1,00,800}{300} = 336 \text{ J K}^{-1} \text{ mol}^{-1}$$

Q52: The wavelength of electrons accelerated from rest through a potential difference of 40 kV is $x \times 10^{-12} m$. The value of x is _____. (Nearest integer)

Given: Mass of electron = $9.1 \times 10^{-31} kg$

Charge on an electron = $1.6 \times 10^{-19} C$

Planck's constant = $6.63 \times 10^{-34} Js$

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Charge on an electron = $1.6 \times 10^{-19} C$

Planck's constant = $6.63 \times 10^{-34} Js$

6.00

Solution:

The kinetic energy of electron after acceleration = $qV = 40000 \times 1.6 \times 10^{-19} = 6.4 \times 10^{-15}$

Using de-Broglie's concept,

$$\lambda = \frac{h}{\sqrt{2m(K.E.)}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 6.4 \times 10^{-15}}} = \frac{6.63 \times 10^{-34}}{\sqrt{116.48 \times 10^{-46}}}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{10.79 \times 10^{-23}} = 6.14 \times 10^{-12}$$

Q53: The vapour pressures of A and B at 25°C are 90 mm Hg and 15 mm Hg respectively. If A and B are mixed such that the mole fraction of A in the mixture is 0.6, then the mole fraction of B in the vapour phase is $x \times 10^{-1}$. The value of x is ____.

(Nearest integer)

Q53: The vapour pressures of A and B at 25°C are 90 mm Hg and 15 mm Hg respectively. If A and B are mixed such that the mole fraction of A in the mixture is 0.6, then the mole fraction of B in the vapour phase is $x \times 10^{-1}$. The value of x is ____.

(Nearest integer)

1.00

Solution:

Since $X_A + X_B = 1 \Rightarrow X_B = 1 - 0.6 = 0.4$

$$\text{Now, } Y_B = \frac{P_B^{\circ} X_B}{P_A^{\circ} X_A + P_B^{\circ} X_B} = \frac{15 \times 0.4}{90 \times 0.6 + 15 \times 0.4} = \frac{6}{54 + 6} = 0.1 = 10^{-1}$$

Q54: 4 g equimolar mixture of NaOH and Na_2CO_3 contains x g of NaOH and y g of Na_2CO_3 . The value of x is _____ g. (Nearest integer)

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1.00

Solution:

Given (i) $x + y = 4$

$$(ii) \frac{x}{40} = \frac{y}{106} \text{ [Equimolar]}$$

$$y = \left[\frac{106}{40} \right] x$$

So $x + \frac{106}{40} x = 4$

$$x + 2.65x = 4$$

$$3.65x = 4$$

$$x = 1.096 \text{ gram.}$$

Q55: When 0.15 g of an organic compound was analyzed using Carius method for estimation of bromine, 0.2397 g of AgBr was obtained. The percentage of bromine in the organic compound is _____. (Nearest integer)

[Atomic mass: Silver = 108, Bromine = 80]

Q55: When 0.15 g of an organic compound was analyzed using Carius method for estimation of bromine, 0.2397 g of AgBr was obtained. The percentage of bromine in the organic compound is _____. (Nearest integer)

[Atomic mass: Silver = 108, Bromine = 80]

68.00

Solution:

188 g of AgBr contains 80 g Br.

Thus, mass of Br present in 0.2397 g AgBr = $\frac{80}{188} \times 0.2397 = 0.102 \text{ g}$

Thus, percentage of Br in organic compound = $\frac{0.102}{0.15} \times 100 = 68\%$

Q56: 100 ml of 0.0018% (w/v) solution of Cl^- ion was the minimum concentration of Cl^- required to precipitate a sol in one hr. The coagulating value of Cl^- ion is _____. Round off your answer up to one decimal place.

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Solution:

Coagulation value is the millimoles of an electrolyte that must be added to 1 L of a colloidal solution for complete coagulation.

For coagulation of 100 mL solution - 0.0018 g Cl^- was required

Thus, for coagulation of 1000 mL solution – 0.018 g Cl^- will be required.

$$\text{Millimoles of } Cl^- = \frac{0.018}{35.5} \times 1000 = 0.507$$



In the above first order reaction the concentration of PCl_5 reduces from initial concentration 50 mol L^{-1} to 10 mol L^{-1} in 120 minutes at 300 K. The rate constant for the reaction at 300 K is $x \times 10^{-2} \text{ min}^{-1}$. The value of x is _____.

[Given $\log 5 = 0.6989$]



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[Given $\log 5 = 0.6989$]

1.34

Solution:

For a first order reaction,

$$\log \frac{[A]_0}{[A]_t} = \frac{kt}{2.303}$$

$$\Rightarrow \log \frac{50}{10} = \frac{k \times 120}{2.303}$$

$$\Rightarrow 0.6989 = \frac{k \times 120}{2.303}$$

$$\Rightarrow k = \frac{0.6989 \times 2.303}{120} = 0.0134$$

$$\Rightarrow k = 1.34 \times 10^{-2}$$

Q58: Diamond has a three-dimensional structure of C atoms formed by covalent bonds. The structure of diamond has face centred cubic lattice where 50% of the tetrahedral voids are also occupied by carbon atoms. The number of carbon atoms present per unit cell of diamond is ____.

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Solution:

Effective number of carbon atoms per unit cell occupying FCC lattice points = 4

Now, we know that number of tetrahedral void sites per unit cell is double of the effective number of atoms, i.e., 8.

Thus, Effective number of carbon atoms per unit cell occupying tetrahedral voids = 50% of 8 = 4

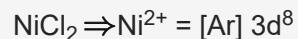
Thus, total number of carbon atoms per unit cell = $4 + 4 = 8$

Q59: An aqueous solution of $NiCl_2$ was heated with excess sodium cyanide in presence of strong oxidizing agent to form $[Ni(CN)_6]^{2-}$. The total change in number of unpaired electrons on metal centre is _____.

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2.00

Solution:



Number of unpaired electrons = 2



Number of unpaired electrons = 0

Thus, difference in unpaired electron = 2

Q60: Potassium chlorate is prepared by electrolysis of KCl in basic solution as shown by following equation.



A current of x A has to be passed for 10 h to produce 10.0 g of potassium chlorate. The value of x is _____. (Nearest integer)

(Molar mass of $KClO_3 = 122.6 \text{ g mol}^{-1}$, $F = 96500 \text{ C}$)

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(Molar mass of $KClO_3 = 122.6 \text{ g mol}^{-1}$, $F = 96500 \text{ C}$)

Solution:

Total charge passed = $It = x \times 10 \times 3600 = 36000x \text{ C}$

Faradays passed = $\frac{36000x}{96500}$

Moles of potassium chlorate formed = $\frac{10}{122.6}$

Now, according to the equation $6OH^- + Cl^- \rightarrow ClO_3^- + 3H_2O + 6e^-$

1 mole of potassium chlorate can be produced by 6 F charge

Thus, $\frac{10}{122.6}$ mole of potassium chlorate will be produced by $6 \times \frac{10}{122.6} \text{ F}$ charge

$$\Rightarrow 6 \times \frac{10}{122.6} = \frac{360x}{965}$$

$$\Rightarrow x = \frac{60 \times 965}{122.6 \times 360} = 1.312 \text{ A}$$

Q61: The lines $x = ay - 1 = z - 2$ and $x = 3y - 2 = bz - 2$, ($ab \neq 0$) are coplanar, if ?

- (A) $b = 1, a \in R - \{0\}$
- (B) $a = 2, b = 3$
- (C) $a = 2, b = 2$
- (D) $a = 1, b \in R - \{0\}$

Q61: The lines $x = ay - 1 = z - 2$ and $x = 3y - 2 = bz - 2$, ($ab \neq 0$) are coplanar, if ?

- (A) $b = 1, a \in R - \{0\}$
- (B) $a = 2, b = 3$
- (C) $a = 2, b = 2$
- (D) $a = 1, b \in R - \{0\}$

Solution:

Lines are $x = ay - 1 = z - 2$

$$\therefore \frac{x}{1} = \frac{y - \frac{1}{a}}{\frac{1}{a}} = \frac{z - 2}{1} \quad \dots (i)$$

and $x = 3y - 2 = bz - 2$

$$\therefore \frac{x}{1} = \frac{y - \frac{2}{3}}{\frac{1}{3}} = \frac{z - \frac{2}{b}}{\frac{1}{b}} \quad \dots (ii)$$

$\therefore$ The lines are co-planar

$$\begin{vmatrix} 0 & -\frac{1}{a} + \frac{2}{3} & -2 + \frac{2}{b} \\ 1 & \frac{1}{a} & 1 \\ 1 & \frac{1}{3} & \frac{1}{b} \end{vmatrix} = 0$$

$$\therefore \begin{vmatrix} 0 & \frac{2}{3} - \frac{1}{a} & \frac{2}{b} - 2 \\ 0 & \frac{1}{a} - \frac{1}{3} & 1 - \frac{1}{b} \\ 1 & \frac{1}{3} & \frac{1}{b} \end{vmatrix} = 0$$

$$\therefore \frac{1}{a} - \frac{1}{ab} = 0$$

$$\Rightarrow b = 1 \text{ and } a \in \mathbb{R} - \{0\}$$

Q62: If the real part of the complex number $(1 - \cos \theta + 2i \sin \theta)^{-1}$ is $\frac{1}{5}$ for $\theta \in (0, \pi)$, then the value of the integral $\int_0^\theta \sin x \, dx$ is equal to ?

- (A) 2
- (B) -1
- (C) 0
- (D) 1

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(A) 2

(B) -1

(C) 0

(D) 1

Solution:

$$\begin{aligned}\text{Let } z &= \frac{1}{1 - \cos \theta + 2i \sin \theta} \\ &= \frac{2\sin^2 \frac{\theta}{2} - 2i \sin \theta}{(1 - \cos \theta)^2 + 4\sin^2 \theta} \\ &= \frac{\sin \frac{\theta}{2} - 2i \cos \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \left(\sin^2 \frac{\theta}{2} + 4\cos^2 \frac{\theta}{2} \right)}\end{aligned}$$

$$\begin{aligned}\operatorname{Re}(z) &= \frac{1}{2 \left(\sin^2 \frac{\theta}{2} + 4\cos^2 \frac{\theta}{2} \right)} = \frac{1}{5} \\ &= \sin^2 \frac{\theta}{2} + 4\cos^2 \frac{\theta}{2} = \frac{5}{2} \\ &= 1 - \cos^2 \frac{\theta}{2} + 4\cos^2 \frac{\theta}{2} = \frac{5}{2} \\ &= 3\cos^2 \frac{\theta}{2} = \frac{3}{2} \\ &= \cos^2 \frac{\theta}{2} = \frac{1}{2}\end{aligned}$$

$$\frac{\theta}{2} = n\pi \pm \frac{\pi}{4}$$

$$\Rightarrow \theta = 2n\pi \pm \frac{\pi}{2}$$

$$\therefore \theta = \frac{\pi}{2} \text{ as } \theta \in (0, \pi)$$

Now,

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin \theta d\theta &= [-\cos \theta]_0^{\frac{\pi}{2}} \\ &= -(0 - 1) = 1.\end{aligned}$$

Q63: In a triangle ABC, if $\left| \overrightarrow{BC} \right| = 3$, $\left| \overrightarrow{CA} \right| = 5$ and $\left| \overrightarrow{BA} \right| = 7$, then the projection of the vector $\overrightarrow{BA}$ on $\overrightarrow{BC}$ is equal to ?

(A) $\frac{11}{2}$

(B) $\frac{13}{2}$

(C) $\frac{19}{2}$

(D) $\frac{15}{2}$

Q63: In a triangle ABC, if $\left| \overrightarrow{BC} \right| = 3$, $\left| \overrightarrow{CA} \right| = 5$ and $\left| \overrightarrow{BA} \right| = 7$, then the projection of the vector $\overrightarrow{BA}$ on $\overrightarrow{BC}$ is equal to ?

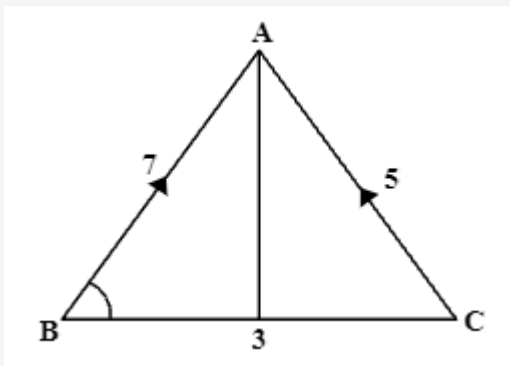
(A) $\frac{11}{2}$

(B) $\frac{13}{2}$

(C) $\frac{19}{2}$

(D) $\frac{15}{2}$

Solution:



$$\begin{aligned}\text{Projection of } \overrightarrow{BA} \text{ on } \overrightarrow{BC} \text{ is equal to} &= \left| \overrightarrow{BA} \cos \angle ABC \right| \\ &= 7 \left| \frac{49+9-25}{2 \cdot 7 \cdot 3} \right| = \frac{11}{2}.\end{aligned}$$

Q64: Let A, B and C be three events such that the probability that exactly one of A and B occurs is $(1 - k)$, the probability that exactly one of B and C occurs is $(1 - 2k)$, the probability that exactly one of C and A occurs is $(1 - k)$ and the probability of all A, B and C occur simultaneously is k^2 , where $0 < k < 1$. Then the probability that at least one of A, B and C occur is ?

- (A) Greater than $\frac{1}{2}$
- (B) Greater than $\frac{1}{4}$ but less than $\frac{1}{2}$
- (C) Exactly equal to $\frac{1}{2}$
- (D) Greater than $\frac{1}{8}$ but less than $\frac{1}{4}$

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- (C) Exactly equal to $\frac{1}{2}$
- (D) Greater than $\frac{1}{8}$ but less than $\frac{1}{4}$

Solution:

$$P(A \cap B \cap C) = k^2$$

$$P(A) + P(B) - 2P(A \cap B) = 1 - k \quad \dots (i)$$

$$P(B) + P(C) - 2P(B \cap C) = 1 - 2k \quad \dots (ii)$$

$$P(C) + P(A) - 2P(A \cap C) = 1 - k \quad \dots (iii)$$

$$(i) + (ii) + (iii)$$

$$P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) = \frac{-4k+3}{2}$$

So,

$$P(A \cup B \cup C) = \frac{-4k+3}{2} + k^2$$

$$P(A \cup B \cup C) = \frac{2k^2 - 4k + 3}{2}$$

$$= \frac{2(k-1)^2 + 1}{2}$$

$$\Rightarrow P(A \cup B \cup C) > \frac{1}{2}.$$

Q65: For the natural numbers m, n , if

$(1 - y)^m (1 + y)^n = 1 + a_1 y + a_2 y^2 + \dots + a_{m+n} y^{m+n}$ and $a_1 = a_2 = 10$, then the value of $(m + n)$ is equal to ?

- (A) 88
- (B) 64
- (C) 100
- (D) 80

Q65: For the natural numbers m, n , if

$(1 - y)^m (1 + y)^n = 1 + a_1 y + a_2 y^2 + \dots + a_{m+n} y^{m+n}$ and $a_1 = a_2 = 10$, then the value of $(m + n)$ is equal to ?

- (A) 88
- (B) 64
- (C) 100
- (D) 80

Solution:

$$\begin{aligned}\text{Coefficient of } y &= 1 \cdot {}^n C_1 + {}^m C_1 (-1) \\ &\Rightarrow n - m = 10 \dots (1)\end{aligned}$$

$$\begin{aligned}\text{Coefficient of } y^2 &= 1 \cdot {}^n C_2 - {}^m C_1 \cdot {}^n C_1 + 1 \cdot {}^m C_2 = 10 \\ &\Rightarrow \frac{n(n-1)}{2} - m \cdot n + \frac{m(m-1)}{2} = 10 \\ &\Rightarrow m^2 + n^2 - 2mn - (n + m) = 20. \\ &\Rightarrow (n - m)^2 - (n + m) = 20 \\ &\therefore (n + m) = 80\end{aligned}$$

Q66: Let r_1 and r_2 be the radii of the largest and smallest circles, respectively, which pass through the point $(-4, 1)$ and having their centres on the circumference of the circle $x^2 + y^2 + 2x + 4y - 4 = 0$. If $\frac{r_1}{r_2} = a + b\sqrt{2}$, then $a + b$ is equal to ?

- (A) 3
- (B) 11
- (C) 5
- (D) 7

Q66: Let r_1 and r_2 be the radii of the largest and smallest circles, respectively, which pass through the point $(-4, 1)$ and having their centres on the circumference of the circle $x^2 + y^2 + 2x + 4y - 4 = 0$. If $\frac{r_1}{r_2} = a + b\sqrt{2}$, then $a + b$ is equal to ?

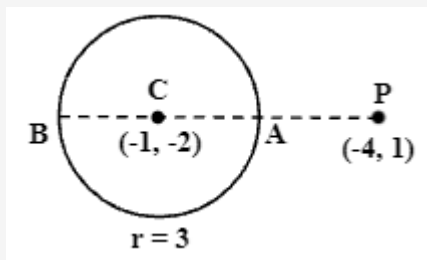
(A) 3

(B) 11

(C) 5

(D) 7

Solution:



Centre of smallest circle is A

Centre of largest circle is B

$$r_2 = |CP - CA| = 3\sqrt{2} - 3$$

$$r_1 = CP + CB = 3\sqrt{2} + 3$$

$$\frac{r_1}{r_2} = \frac{(3\sqrt{2}+3)^2}{9} = (\sqrt{2} + 1)^2 = 3 + 2\sqrt{2}$$

$$\Rightarrow a = 3, b = 2$$

$$\therefore a + b = 5$$

Q67: The value of $\tan\left(2\tan^{-1}\left(\frac{3}{5}\right) + \sin^{-1}\left(\frac{5}{13}\right)\right)$ is equal to ?

(A) $\frac{-291}{76}$

(B) $\frac{-181}{69}$

(C) $\frac{151}{63}$

(D) $\frac{220}{21}$

Q67: The value of $\tan\left(2\tan^{-1}\left(\frac{3}{5}\right) + \sin^{-1}\left(\frac{5}{13}\right)\right)$ is equal to ?

(A) $\frac{-291}{76}$

(B) $\frac{-181}{69}$

(C) $\frac{151}{63}$

(D) $\frac{220}{21}$

Solution:

Consider

$$\begin{aligned} & \left(2\tan^{-1}\left(\frac{3}{5}\right) + \sin^{-1}\left(\frac{5}{13}\right) \right) \\ &= \underbrace{\tan^{-1}\frac{3}{5} + \tan^{-1}\frac{3}{5}}_{x>0, y>0, xy<1} + \tan^{-1}\frac{5}{12} \\ &= \tan^{-1}\frac{\frac{6}{5}}{1-\frac{9}{25}} + \tan^{-1}\frac{5}{12} \\ &= \underbrace{\tan^{-1}\frac{15}{8} + \tan^{-1}\frac{5}{12}}_{x>0, y>0, xy<1} \\ &= \tan^{-1}\frac{\frac{15}{8} + \frac{5}{12}}{1 - \frac{15}{8} \cdot \frac{5}{12}} = \tan^{-1}\frac{220}{21} \end{aligned}$$

Now,

$$\begin{aligned} & \tan\left(2\tan^{-1}\left(\frac{3}{5}\right) + \sin^{-1}\left(\frac{5}{13}\right)\right) \\ & \therefore \tan\left(\tan^{-1}\frac{220}{21}\right) = \frac{220}{21}. \end{aligned}$$

Q68: If the mean and variance of six observations 7, 10, 11, 15, a , b are 10 and $\frac{20}{3}$, respectively, then the value of $|a - b|$ is equal to ?

(A) 7

(B) 11

(C) 9

(D) 1

Q68: If the mean and variance of six observations 7, 10, 11, 15, a , b are 10 and $\frac{20}{3}$, respectively, then the value of $|a - b|$ is equal to ?

(A) 7

(B) 11

(C) 9

(D) 1

Solution:

Given,

$$10 = \frac{7+10+11+15+a+b}{6}$$

$$\Rightarrow a + b = 17 \dots\dots(i)$$

and

$$\frac{20}{3} = \frac{7^2+10^2+11^2+15^2+a^2+b^2}{6} - 10^2$$

$$\Rightarrow a^2 + b^2 = 145 \dots\dots(ii)$$

By Solving (i) and (ii), we get

$$a = 9, b = 8 \text{ or } a = 8, b = 9$$

$$\therefore |a - b| = 1.$$

Q69: If sum of the first 21 terms of the series $\log_{9^{1/2}} x + \log_{9^{1/3}} x + \log_{9^{1/4}} x + \dots$, where $x > 0$ is 504, then x is equal to ?

- (A) 81
- (B) 243
- (C) 7
- (D) 9

Q69: If sum of the first 21 terms of the series $\log_{9^{1/2}} x + \log_{9^{1/3}} x + \log_{9^{1/4}} x + \dots$, where $x > 0$ is 504, then x is equal to ?

(A) 81

(B) 243

(C) 7

(D) 9

Solution:

Given, $\log_{9^{1/2}} x + \log_{9^{1/3}} x + \log_{9^{1/4}} x + \dots$

Let $S = 2\log_9 x + 3\log_9 x + \dots + 22\log_9 x$

$$S = \log_9 x (2 + 3 + \dots + 22)$$

$$S = \log_9 x \left(\frac{21}{2} (2 + 22) \right)$$

$$S = 252\log_9 x$$

Given,

$$252\log_9 x = 504$$

$$\Rightarrow \log_9 x = 2 \Rightarrow x = 81.$$

Q70: Let in a right-angled triangle, the smallest angle be θ . If a triangle formed by taking reciprocal of its sides is also a right angled triangle, then $\sin \theta$ is equal to ?

(A) $\frac{\sqrt{5}+1}{4}$

(B) $\frac{\sqrt{5}-1}{2}$

(C) $\frac{\sqrt{2}-1}{2}$

(D) $\frac{\sqrt{5}-1}{4}$

Q70: Let in a right-angled triangle, the smallest angle be θ . If a triangle formed by taking reciprocal of its sides is also a right angled triangle, then $\sin \theta$ is equal to ?

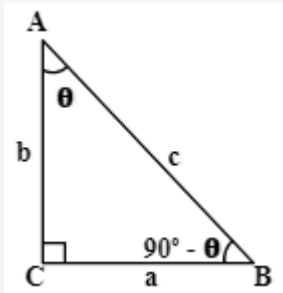
(A) $\frac{\sqrt{5}+1}{4}$

(B) $\frac{\sqrt{5}-1}{2}$

(C) $\frac{\sqrt{2}-1}{2}$

(D) $\frac{\sqrt{5}-1}{4}$

Solution:



$$\angle A = \theta$$

$$\angle B = 90^\circ - \theta$$

a is smallest side

$$c^2 = a^2 + b^2$$

$$\therefore \frac{1}{a^2} = \frac{1}{b^2} + \frac{1}{c^2}$$

$$\frac{b^2 c^2}{a^2} = b^2 + c^2$$

$$\text{Use, } a = 2R \sin A = 2R \sin \theta$$

$$b = 2R \sin B = 2R \sin(90^\circ - \theta) = 2R \cos \theta$$

$$c = 2R \sin C = 2 \sin 90^\circ = 2R$$

$$\therefore \frac{4R^2 \cos^2 \theta \cdot 4R^2}{4R^2 \sin^2 \theta} = 4R^2 \cos^2 \theta + 4R^2$$

$$\cos^2 \theta = \sin^2 \theta \cos^2 \theta + \sin^2 \theta$$

$$1 - \sin^2 \theta = \sin^2 \theta (1 - \sin^2 \theta) + \sin^2 \theta$$

$$\sin^2 \theta = \frac{3 - \sqrt{5}}{2}$$

$$\Rightarrow \sin \theta = \frac{\sqrt{5} - 1}{2}$$

Q71: Let $f : R - \left\{ \frac{\alpha}{6} \right\} \rightarrow R$ be defined by $f(x) = \frac{5x+3}{6x-\alpha}$. Then the value of α for which $(f \circ f)(x) = x$, for all $x \in R - \left\{ \frac{\alpha}{6} \right\}$, is ?

- (A) No such α exists
- (B) 5
- (C) 6
- (D) 8

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(A) No such α exists

(B) 5

(C) 6

(D) 8

Solution:

Let

$$f(x) = \frac{5x+3}{6x-\alpha} = y \quad \dots (i)$$

$$\Rightarrow 5x + 3 = 6xy - \alpha y$$

$$\Rightarrow x(6y - 5) = \alpha y + 3$$

$$\therefore x = \frac{\alpha y + 3}{6y - 5}$$

$$\Rightarrow f^{-1}(x) = \frac{\alpha x + 3}{6x - 5} \quad \dots (ii)$$

Now,

$$f \circ f(x) = x$$

$$\Rightarrow f(x) = f^{-1}(x)$$

From equation (i) and (ii)

Clearly, $\alpha = 5$.

Q72: Let P be a variable point on the parabola $y = 4x^2 + 1$. Then the locus of the mid-point of the point P and the foot of the perpendicular drawn from the point P to the line $y = x$ is ?

- (A) $(3x - y)^2 + (x - 3y) + 2 = 0$
- (B) $2(x - 3y)^2 + (3x - y) + 2 = 0$
- (C) $2(3x - y)^2 + (x - 3y) + 2 = 0$
- (D) $(3x - y)^2 + 2(x - 3y) + 2 = 0$

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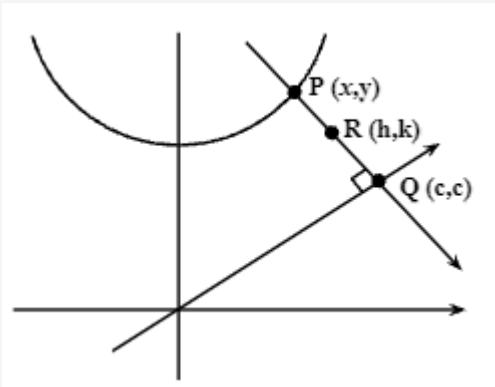
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(C) $2(3x - y)^2 + (x - 3y) + 2 = 0$

(D) $(3x - y)^2 + 2(x - 3y) + 2 = 0$

Solution:



$$\frac{k-c}{h-c} = -1$$

$$\Rightarrow c = \frac{h+k}{2} \text{ and } P(x, y)$$

$$R = \left(\frac{x+c}{2}, \frac{y+c}{2} \right)$$

$$R = \left(\frac{x}{2} + \frac{h}{4} + \frac{k}{4}, \frac{y}{2} + \frac{h}{4} + \frac{k}{4} \right)$$

$$\therefore h = \frac{x}{2} + \frac{h}{4} + \frac{k}{4} \text{ and } k = \frac{y}{2} + \frac{h}{4} + \frac{k}{4}$$

$$\Rightarrow x = \frac{3h}{2} - \frac{k}{2}, \quad y = \frac{3k}{2} - \frac{h}{2},$$

Now,

$$y = 4x^2 + 1$$

$$\left(\frac{3k-h}{2} \right) = 4 \left(\frac{3h-k}{2} \right)^2 + 1$$

$$\Rightarrow 2(3h-k)^2 + (h-3k) + 2 = 0$$

Q73: If $[x]$ denotes the greater integer less than or equal to x , then the value of the integral

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [[x] - \sin x] dx \text{ is equal to ?}$$

- (A) 0
- (B) π
- (C) 1
- (D) $-\pi$

Q73: If $[x]$ denotes the greater integer less than or equal to x , then the value of the integral

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [[x] - \sin x] dx \text{ is equal to ?}$$

(A) 0

(B) π

(C) 1

(D) $-\pi$

Solution:

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} ([x] + [-\sin x]) dx \dots\dots (i)$$

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} ([-x] + [\sin x]) dx$$

By property

$$2I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\left(\underbrace{[x] + [-x]}_{-1} \right) + \left(\underbrace{[\sin x] + [-\sin x]}_{-1} \right) \right) dx$$

$$2I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (-2) dx = -2(\pi)$$

$$I = -\pi.$$

Q74: Let $y = y(x)$ satisfies the equation $\frac{dy}{dx} - |A| = 0$, for all $x > 0$, where

$$A = \begin{bmatrix} y & \sin x & 1 \\ 0 & -1 & 1 \\ 2 & 0 & \frac{1}{x} \end{bmatrix}. \text{ If } y(\pi) = \pi + 2, \text{ then the value of } y\left(\frac{\pi}{2}\right) \text{ is ?}$$

(A) $\frac{\pi}{2} - \frac{4}{\pi}$

(B) $\frac{3\pi}{2} - \frac{1}{\pi}$

(C) $\frac{\pi}{2} - \frac{1}{\pi}$

(D) $\frac{\pi}{2} + \frac{4}{\pi}$

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(B) $\frac{3\pi}{2} - \frac{1}{\pi}$

(C) $\frac{\pi}{2} - \frac{1}{\pi}$

(D) $\frac{\pi}{2} + \frac{4}{\pi}$

Solution:

$$|A| = \frac{-y}{x} + 2 \sin x + 2$$

$$\frac{dy}{dx} = |A|$$

$$\frac{dy}{dx} = \frac{-y}{x} + 2 \sin x + 2$$

$$\frac{dy}{dx} + \frac{y}{x} = 2 \sin x + 2$$

$$I.F. = e^{\int \frac{1}{x} dx} = x$$

$$\Rightarrow yx = \int x (2 \sin x + 2) dx$$

$$xy = x^2 - 2x \cos x + 2 \sin x + c \dots\dots (i)$$

Now, $x = \pi, y = \pi + 2$ in equation (i)

$$\therefore c = 0$$

Now the equation (i) becomes

$$xy = x^2 - 2x \cos x + 2 \sin x$$

$$\text{put } x = \frac{\pi}{2}$$

$$\Rightarrow \frac{\pi}{2} y = \frac{\pi^2}{4} + 2$$

$$\therefore y = \frac{\pi}{2} + \frac{4}{\pi}$$

Q75: The sum of all the local minimum values of the twice differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3 - 3x^2 - \frac{3f''(2)}{2}x + f''(1)$ is ?

(A) -22

(B) 0

(C) -27

(D) 5

Q75: The sum of all the local minimum values of the twice differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3 - 3x^2 - \frac{3f''(2)}{2}x + f''(1)$ is ?

(A) -22

(B) 0

(C) -27

(D) 5

Solution:

$$f(x) = x^3 - 3x^2 - \frac{3}{2}f''(2)x + f''(1) \quad \dots (i)$$

$$f'(x) = 3x^2 - 6x - \frac{3}{2}f''(2) \quad \dots (ii)$$

$$f''(x) = 6x - 6 \quad \dots (iii)$$

$$f''(2) = 12 - 6 = 6$$

$$\text{and } f''(1) = 0$$

Using equation (ii)

$$f'(x) = 3x^2 - 6x - \frac{3}{2}f''(2)$$

$$f'(x) = 3x^2 - 6x - \frac{3}{2} \times 6$$

$$f'(x) = 3x^2 - 6x - 9$$

Now,

$$f'(x) = 0$$

$$3x^2 - 6x - 9 = 0$$

$$\Rightarrow x = -1 \text{ and } \Rightarrow x = 3$$

Using equation (iii)

$$f''(x) = 6x - 6$$

$$f''(-1) = -12 < 0 \text{ maxima}$$

$$f''(3) = 12 > 0 \text{ minima}$$

Using equation (i)

$$f(x) = x^3 - 3x^2 - \frac{3}{2}f''(2)x + f''(1)$$

$$f(x) = x^3 - 3x^2 - \frac{3}{2} \times 6x + 0$$

$$f(x) = x^3 - 3x^2 - 9x$$

$$f(3) = 27 - 27 - 9 \times 3 = -27 .$$

Q76: The value of $k \in \mathcal{R}$, for which the following system of linear equations

$$3x - y + 4z = 3,$$

$$x + 2y - 3z = -2,$$

$$6x + 5y + kz = -3,$$

has infinitely many solutions, is:

(A) 3

(B) -3

(C) 5

(D) -5

Q76: The value of $k \in \mathbb{R}$, for which the following system of linear equations

$$3x - y + 4z = 3,$$

$$x + 2y - 3z = -2,$$

$$6x + 5y + kz = -3,$$

has infinitely many solutions, is:

(A) 3

(B) -3

(C) 5

(D) -5

Solution:

We have, $\Delta = \Delta_x = \Delta_y = \Delta_z = 0$

$$\begin{vmatrix} 3 & -1 & 4 \\ 1 & 2 & -3 \\ 6 & 5 & k \end{vmatrix} = 0$$

$$\Rightarrow 3(2k + 15) + k + 18 - 28 = 0$$

$$\Rightarrow 7k + 35 = 0 \Rightarrow k = -5$$

Q77: Consider the following three statements:

(A) If $3 + 3 = 7$ then $4 + 3 = 8$.

(B) If $5 + 3 = 8$ then earth is flat.

(C) If both (A) and (B) are true then $5 + 6 = 17$.

Then, which of the following statements is correct?

(A) (A) and (C) are true while (B) is false

(B) (A) is true while (B) and (C) are false

(C) (A) is false, but (B) and (C) are true

(D) (A) and (B) are false while (C) is true

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(C) (A) is false, but (B) and (C) are true

(D) (A) and (B) are false while (C) is true

Solution:

Truth Table

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Q78: Let $g(t) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos\left(\frac{\pi}{4}t + f(x)\right) dx$, where $f(x) = \log_e\left(x + \sqrt{x^2 + 1}\right)$, $x \in R$.

Then which one of the following is correct ?

(A) $g(1) + g(0) = 0$

(B) $g(1) = \sqrt{2}g(0)$

(C) $g(1) = g(0)$

(D) $\sqrt{2}g(1) = g(0)$

Q78: Let $g(t) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos\left(\frac{\pi}{4}t + f(x)\right) dx$, where $f(x) = \log_e\left(x + \sqrt{x^2 + 1}\right)$, $x \in R$.

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(B) $g(1) = \sqrt{2}g(0)$

(C) $g(1) = g(0)$

(D) $\sqrt{2}g(1) = g(0)$

Solution:

$$g(t) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos\left(\frac{\pi}{4}t + f(x)\right) dx$$

$$g(1) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos\left(\frac{\pi}{4} + f(x)\right) dx$$

$$g(1) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \frac{\pi}{4} \cos(f(x)) dx - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin \frac{\pi}{4} \sin(f(x)) dx$$

$$g(1) = \frac{1}{\sqrt{2}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(f(x)) dx - \frac{1}{\sqrt{2}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin(\log(x + \sqrt{x^2 + 1})) dx$$

$$g(1) = \frac{1}{\sqrt{2}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(f(x)) dx + 0$$

$$g(1) = \frac{1}{\sqrt{2}} g(0)$$

$$\therefore \sqrt{2}g(1) = g(0)$$

Q79: Consider the line L given by the equation $\frac{x-3}{2} = \frac{y-1}{1} = \frac{z-2}{1}$. Let Q be the mirror image of the point $(2, 3, -1)$ with respect to L. Let a plane P be such that it passes through Q, and the line L is perpendicular to P. Then which of the following points is on the plane P ?

- (A) $(1, 2, 2)$
- (B) $(-1, 1, 2)$
- (C) $(1, 1, 1)$
- (D) $(1, 1, 2)$

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(A) $(1, 2, 2)$

(B) $(-1, 1, 2)$

(C) $(1, 1, 1)$

(D) $(1, 1, 2)$

Solution:

Plane P is $\perp^r$ to the line $\frac{x-3}{2} = \frac{y-1}{1} = \frac{z-2}{1}$

& Passes through the point $(2, 3, -1)$

Then the equation of plane P is

$$2(x - 2) + 1(y - 3) + 1(z + 1) = 0$$

$$2x + y + z - 6 = 0$$

The point $(1, 2, 2)$ satisfies the above equation.

Q80: If $f : R \rightarrow R$ is given by $f(x) = x + 1$, then the value of

$\lim_{n \rightarrow \infty} \frac{1}{n} \left[f(0) + f\left(\frac{5}{n}\right) + f\left(\frac{10}{n}\right) + \dots + f\left(\frac{5(n-1)}{n}\right) \right]$, is:

(A) $\frac{3}{2}$

(B) $\frac{7}{2}$

(C) $\frac{5}{2}$

(D) $\frac{1}{2}$

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(B) $\frac{7}{2}$

(C) $\frac{5}{2}$

(D) $\frac{1}{2}$

Solution:

$$I = \sum_{r=0}^{n-1} f\left(\frac{5r}{n}\right) \frac{1}{n}$$

$$I = \int_0^1 f(5x) dx$$

$$I = \int_0^1 (5x + 1) dx$$

$$I = \left(\frac{5x^2}{2} + x \right)_0^1$$

$$I = \frac{5}{2} + 1 = \frac{7}{2}$$

Q81: Let a curve $y = y(x)$ be given by the solution of the differential equation

$$\cos\left(\frac{1}{2}\cos^{-1}(e^{-x})\right)dx = \sqrt{e^{2x} - 1}dy.$$

If it intersects y-axis at $y = -1$, and the intersection point of the curve with x-axis is $(\alpha, 0)$ the e^α is equal to _____ ?

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If it intersects y-axis at $y = -1$, and the intersection point of the curve with x-axis is $(\alpha, 0)$ the e^α is equal to _____ ?

Solution:

$$\cos\left(\frac{1}{2}\cos^{-1}(e^{-x})\right)dx = \sqrt{e^{2x}-1} dy$$

$$\text{Put } \cos^{-1}(e^{-x}) = \theta, \quad \theta \in [0, \pi]$$

$$\cos \theta = e^{-x} \Rightarrow 2\cos^2 \frac{\theta}{2} - 1 = e^{-x}$$

$$\cos \frac{\theta}{2} = \sqrt{\frac{e^{-x}+1}{2}} = \sqrt{\frac{e^x+1}{2e^x}}$$

$$\sqrt{\frac{e^x+1}{2e^x}} dx = \sqrt{e^{2x}-1} dy$$

$$\frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{e^x}\sqrt{e^x-1}} = \int dy$$

$$\text{Put } e^x = t$$

$$\Rightarrow \frac{dt}{dx} = e^x$$

$$\int \frac{dt}{t\sqrt{t^2-t}} = \sqrt{2}y$$

$$\text{Put } t = \frac{1}{z} \Rightarrow \frac{dt}{dz} = -\frac{1}{z^2}$$

$$\int \frac{-\frac{dz}{z^2}}{\frac{1}{z}\sqrt{\frac{1}{z^2}-\frac{1}{z}}} = \sqrt{2}y.$$

$$-\int \frac{dz}{\sqrt{1-z}} = \sqrt{2}y$$

$$\frac{-2(1-z)^{\frac{1}{2}}}{-1} = \sqrt{2}y + c$$

$$2\left(1 - \frac{1}{t}\right)^{\frac{1}{2}} = \sqrt{2}y + c$$

$$2(1 - e^{-x})^{\frac{1}{2}} = \sqrt{2}y + c$$

By the given condition.

$$\therefore c = \sqrt{2}$$

$$2(1 - e^{-x})^{\frac{1}{2}} = \sqrt{2}(y + 1),$$

passes through $(\alpha, 0)$

$$2(1 - e^{-\alpha})^{\frac{1}{2}} = \sqrt{2}$$

$$\sqrt{1 - e^{-\alpha}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow 1 - e^{-\alpha} = \frac{1}{2}$$

$$e^{-\alpha} = \frac{1}{2}$$

$$\therefore e^{\alpha} = 2$$

Q82: For $k \in N$, let $\frac{1}{\alpha(\alpha+1)(\alpha+2)\dots(\alpha+20)} = \sum_{k=0}^{20} \frac{A_k}{\alpha+k}$, where $\alpha > 0$. Then the value of

$100 \left(\frac{A_{14} + A_{15}}{A_{13}} \right)^2$ is equal to _____ ?

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$100 \left(\frac{A_{14} + A_{15}}{A_{13}} \right)^2$ is equal to _____ ?

9.00

Solution:

Given,

$$\frac{1}{\alpha(\alpha+1)(\alpha+2)\dots(\alpha+20)} = \sum_{k=0}^{20} \frac{A_k}{\alpha+k}$$

$$A_{14} = \frac{1}{(-14)(-13)\dots(-1)(1)(2)\dots(6)} = \frac{1}{14!6!}$$

$$A_{15} = \frac{1}{(-15)(-14)\dots(-1)(1)(2)\dots(5)} = \frac{-1}{15!5!}$$

$$A_{13} = \frac{1}{(-13)\dots(-1)(1)(2)\dots(7)} = \frac{-1}{13!7!}$$

$$\frac{A_{14}}{A_{13}} = \frac{-7}{14} = -\frac{1}{2}$$

$$\frac{A_{15}}{A_{13}} = \frac{42}{15 \times 14} = \frac{1}{5}$$

$$100 \left(\frac{A_{14}}{A_{13}} + \frac{A_{15}}{A_{13}} \right)^2 = 100 \left(-\frac{1}{2} + \frac{1}{5} \right)^2 = 9.$$

Q83: Let a function $g : [0, 4] \rightarrow \mathcal{R}$ be defined as,

$$g(x) = \begin{cases} \max \{t^3 - 6t^2 + 9t - 3\}, & 0 \leq t \leq x, 0 \leq x \leq 3 \\ 4 - x, & 3 < x \leq 4 \end{cases}, \text{ then the number of}$$

points in the interval $(0, 4)$ where $g(x)$ is NOT differentiable, is _____ .

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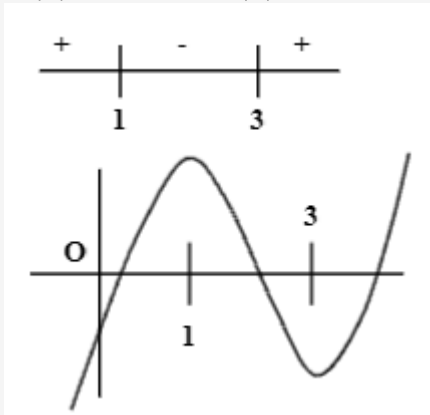
points in the interval $(0, 4)$ where $g(x)$ is NOT differentiable, is _____ .

Solution:

$$f(x) = x^3 - 6x^2 + 9x - 3$$

$$f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$$

$$f(1) = 1 \text{ and } f(3) = -3$$



$$g(x) = \begin{cases} f(x), & 0 \leq x \leq 1 \\ 0, & 1 \leq x \leq 3 \\ 4-x, & 3 < x \leq 4 \end{cases}$$

$g(x)$ is continuous

$$g'(x) = \begin{cases} 3(x-1)(x-3), & 0 \leq x \leq 1 \\ 0, & 1 \leq x \leq 3 \\ -1, & 3 < x \leq 4 \end{cases}$$

$g(x)$ is non-differentiable at $x = 3$.

Q84: Let $A = \{a_{ij}\}$ be a 3×3 matrix, where $a_{ij} = \begin{cases} (-1)^{j-i} & \text{if } i < j, \\ 2 & \text{if } i = j, \\ (-1)^{i+j} & \text{if } i > j \end{cases}$

then $\det(3Adj(2A^{-1}))$ is equal to _____ ?

Q84: Let $A = \{a_{ij}\}$ be a 3×3 matrix, where $a_{ij} = \begin{cases} (-1)^{j-i} & \text{if } i < j, \\ 2 & \text{if } i = j, \\ (-1)^{i+j} & \text{if } i > j \end{cases}$

then $\det(3\text{Adj}(2A^{-1}))$ is equal to _____ ?

108.00

Solution:

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

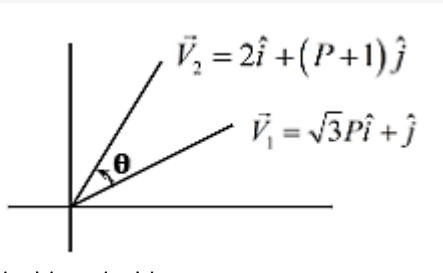
$$|A| = 4$$

$$\begin{aligned} |3\text{adj}(2A^{-1})| &= |3 \cdot 2^2 \text{adj}(A^{-1})| \\ &= 12^3 |\text{adj}(A^{-1})| \\ &= 12^3 |A^{-1}|^2 = \frac{12^3}{|A|^2} = \frac{12^3}{16} = 108. \end{aligned}$$

Q85: For $p > 0$, a vector $\vec{v}_2 = 2\hat{i} + (p + 1)\hat{j}$ is obtained by rotating the vector $\vec{v}_1 = \sqrt{3}p\hat{i} + \hat{j}$ by an angle θ about the origin in a counter clockwise direction. If $\tan \theta = \frac{(\alpha\sqrt{3}-2)}{4\sqrt{3}+3}$, then the value of α is equal to ?

Q85: For $p > 0$, a vector $\vec{v}_2 = 2\hat{i} + (p + 1)\hat{j}$ is obtained by rotating the vector $\vec{v}_1 = \sqrt{3}p\hat{i} + \hat{j}$ by an angle θ about the origin in a counter clockwise direction. If $\tan \theta = \frac{(\alpha\sqrt{3}-2)}{4\sqrt{3}+3}$, then the value of α is equal to ?

Solution:



$$|\vec{V}_1| = |\vec{V}_2|$$

$$3P^2 + 1 = 4 + (P + 1)^2$$

$$2P^2 - 2P - 4 = 0$$

$$\Rightarrow P^2 - P - 2 = 0$$

$$P = 2 \text{ and } P = -1 \text{ (rejected)}$$

$$\cos \theta = \frac{\vec{V}_1 \cdot \vec{V}_2}{|\vec{V}_1| |\vec{V}_2|}$$

$$\cos \theta = \frac{4\sqrt{3}+3}{\sqrt{13}\sqrt{13}} = \frac{4\sqrt{3}+3}{13}$$

$$\tan \theta = \frac{\sqrt{112-24\sqrt{3}}}{4\sqrt{3}+\sqrt{3}} = \frac{6\sqrt{3}-2}{4\sqrt{3}+3} = \frac{\alpha\sqrt{3}-2}{4\sqrt{3}+3}$$

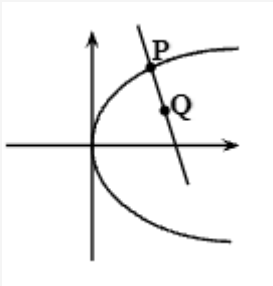
$$\Rightarrow \alpha = 6.$$

Q86: If the point on the curve $y^2 = 6x$, nearest to the point $(3, \frac{3}{2})$ is (α, β) , then $2(\alpha + \beta)$ is equal to _____ ?

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9.00

Solution:



Minimum distance is along the normal

$$P \equiv \left(\frac{3}{2}t^2, 3t\right)$$

Normal at point P

$$tx + y = 3t + \frac{3}{2}t^3$$

Passes through $(3, \frac{3}{2})$

$$\Rightarrow 3t + \frac{3}{2} = 3t + \frac{3}{2}t^3$$

$$\Rightarrow t^3 = 1$$

$$\therefore t = 1$$

$$P \equiv \left(\frac{3}{2}t, 3t^2\right) = (\alpha, \beta)$$

$$2(\alpha + \beta) = 2\left(\frac{3}{2} + 3\right) = 9.$$

Q87: If $\lim_{x \rightarrow 0} \frac{\alpha x e^x - \beta \log_e(1+x) + \gamma x^2 e^{-x}}{x \sin^2 x} = 10$, $\alpha, \beta, \gamma \in \mathbb{R}$, then the value of $\alpha + \beta + \gamma$ is

?

Q87: If $\lim_{x \rightarrow 0} \frac{\alpha x e^x - \beta \log_e(1+x) + \gamma x^2 e^{-x}}{x \sin^2 x} = 10$, $\alpha, \beta, \gamma \in \mathbb{R}$, then the value of $\alpha + \beta + \gamma$ is

?

3.00

Solution:

We have,

$$\lim_{x \rightarrow 0} \frac{\alpha x e^x - \beta \log_e(1+x) + \gamma x^2 e^{-x}}{x \sin^2 x} = 10$$

$$\lim_{x \rightarrow 0} \frac{\alpha x \left(1 + x + \frac{x^2}{2}\right) - \beta \left(x - \frac{x^2}{2} + \frac{x^3}{3}\right) + \gamma x^2 (1-x)}{x^3} = 10$$

$$\lim_{x \rightarrow 0} \frac{x(\alpha - \beta) + x^2 \left(\alpha + \frac{\beta}{2} + \gamma\right) + x^3 \left(\frac{\alpha}{2} - \frac{\beta}{3} - \gamma\right)}{x^3} = 10$$

For limit to exist

$$\alpha - \beta = 0, \alpha + \frac{\beta}{2} + \gamma = 0$$

$$\therefore \frac{\alpha}{2} - \frac{\beta}{3} - \gamma = 10 \dots (i)$$

$$\beta = \alpha, \gamma = -\frac{3\alpha}{2}$$

Put in (i), we get

$$\frac{\alpha}{2} - \frac{\alpha}{3} + \frac{3\alpha}{2} = 10$$

$$\frac{\alpha}{6} + \frac{3\alpha}{2} = 10 \Rightarrow \frac{\alpha + 9\alpha}{6} = 10$$

$$\Rightarrow \alpha = 6$$

$$\alpha = 6, \beta = 6 \text{ and } \gamma = -9$$

$$\therefore \alpha + \beta + \gamma = 3$$

Q88: The number of solutions of the equation

$$\log_{(x+1)} (2x^2 + 7x + 5) + \log_{(2x+5)} (x + 1)^2 - 4 = 0, x > 0, \text{ is } \underline{\hspace{2cm}} ?$$

Q88: The number of solutions of the equation

$$\log_{(x+1)} (2x^2 + 7x + 5) + \log_{(2x+5)} (x+1)^2 - 4 = 0, x > 0, \text{ is } \underline{\hspace{2cm}} ?$$

1.00

Solution:

$$\log_{(x+1)} (2x^2 + 7x + 5) + \log_{(2x+5)} (x+1)^2 - 4 = 0$$

$$\log_{(x+1)} (2x+5) (x+1) + 2\log_{(2x+5)} (x+1) = 4$$

$$\log_{(x+1)} (2x+5) + 1 + 2\log_{(2x+5)} (x+1) = 4$$

$$\text{Put } \log_{(x+1)} (2x+5) = t$$

$$\Rightarrow t + \frac{2}{t} = 3$$

$$\Rightarrow t^2 - 3t + 2 = 0$$

$$\Rightarrow t = 1, 2$$

$$\therefore \log_{(x+1)} (2x+5) = 1 \text{ \& } \log_{(x+1)} (2x+5) = 2$$

$$x+1 = 2x+5 \text{ \& } 2x+5 = (x+1)^2$$

$$x = -4 \text{ (rejected) and } x^2 = 4 \Rightarrow x = 2, x = -2 \text{ (rejected)}$$

$$\text{So, } x = 2$$

$\therefore$ The number of solutions = 1.

Q89: Let $\{a_n\}_{n=1}^{\infty}$ be a sequence such that $a_1 = 1, a_2 = 1$ and $a_{n+2} = 2a_{n+1} + a_n$ for all $n \geq 1$. Then the value of $47 \sum_{n=1}^{\infty} \frac{a_n}{2^{3n}}$ is equal to _____ ?

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7.00

Solution:

$$\text{Let } \sum_{n=1}^{\infty} \frac{a_n}{8^n} = P$$

$$a_{n+2} = 2a_{n+1} + a_n$$

Divide by 8^n on both sides, we get

$$\frac{a_{n+2}}{8^n} = \frac{2a_{n+1}}{8^n} + \frac{a_n}{8^n}$$

$$\Rightarrow 64 \frac{a_{n+2}}{8^{n+2}} = \frac{16a_{n+1}}{8^{n+1}} + \frac{a_n}{8^n}$$

$$64 \sum_{n=1}^{\infty} \frac{a_{n+2}}{8^{n+2}} = 16 \sum_{n=1}^{\infty} \frac{a_{n+1}}{8^{n+1}} + \sum_{n=1}^{\infty} \frac{a_n}{8^n}$$

$$64 \left(P - \frac{a_1}{8} - \frac{a_2}{8^2} \right) = 16 \left(P - \frac{a_1}{8} \right) + P$$

$$\Rightarrow 64 \left(P - \frac{1}{8} - \frac{1}{64} \right) = 16 \left(P - \frac{1}{8} \right) + P$$

$$\Rightarrow 64P - 8 - 1 = 16P - 2 + P$$

$$\Rightarrow 47P = 7$$

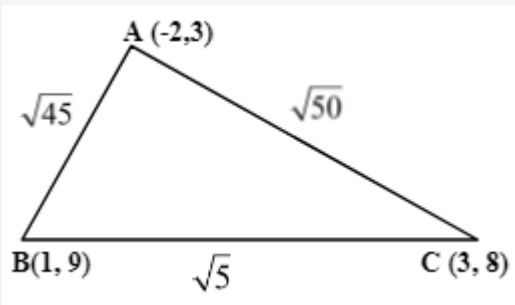
$$\therefore 47 \sum_{n=1}^{\infty} \frac{a_n}{2^{3n}} = 7$$

Q90: Consider a triangle having vertices $A(-2, 3)$, $B(1, 9)$ and $C(3, 8)$. If a line L passing through the circumcentre of triangle ABC , bisects line BC , and intersects y -axis at point $(0, \frac{\alpha}{2})$, then the value of real number α is _____ ?

Q90: Consider a triangle having vertices $A(-2, 3)$, $B(1, 9)$ and $C(3, 8)$. If a line L passing through the circumcentre of triangle ABC , bisects line BC , and intersects y -axis at point $(0, \frac{\alpha}{2})$, then the value of real number α is _____ ?

9.00

Solution:



In triangle ABC

$$(AC)^2 = (AB)^2 + (BC)^2$$

$$(\sqrt{50})^2 = (\sqrt{45})^2 + (\sqrt{5})^2$$

$$\therefore \angle B = 90^\circ$$

$$\text{Circum-center} = \left(\frac{1}{2}, \frac{11}{2}\right)$$

$$\text{Mid point of } BC = \left(2, \frac{17}{2}\right)$$

$$\text{Line : } \left(y - \frac{11}{2}\right) = 2\left(x - \frac{1}{2}\right)$$

$$\Rightarrow y = 2x + \frac{9}{2}$$

$$\text{Since, the line } y = 2x + \frac{9}{2} \text{ passing through } \left(0, \frac{\alpha}{2}\right)$$

$$\therefore \frac{\alpha}{2} = \frac{9}{2} \Rightarrow \alpha = 9.$$